3D and 4D World Modeling: A Survey

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World modeling has become a cornerstone in AI research, enabling agents to understand, represent, and predict the dynamic environments they inhabit. While prior work largely emphasizes generative methods for 2D image and video data, they overlook the rapidly growing body of work that leverages **native 3D and 4D representations** such as RGB-D imagery, occupancy grids, and LiDAR point clouds for large-scale scene modeling. At the same time, the absence of a standardized definition and taxonomy for "world models" has led to fragmented and sometimes inconsistent claims in the literature. This survey addresses these gaps by presenting the first comprehensive review explicitly dedicated to 3D and 4D world modeling and generation. We establish precise definitions, introduce a structured taxonomy spanning video-based (**VideoGen**), occupancy-based (**OccGen**), and LiDAR-based (**LiDARGen**) approaches, and systematically summarize datasets and evaluation metrics tailored to 3D/4D settings. We further discuss practical applications, identify open challenges, and highlight promising research directions, aiming to provide a coherent and foundational reference for advancing the field. A systematic summary of existing literature is available on our project page.

Project Page: https://worldbench.github.io/survey

GitHub Repo: https://github.com/worldbench/awesome-3d-4d-world-models

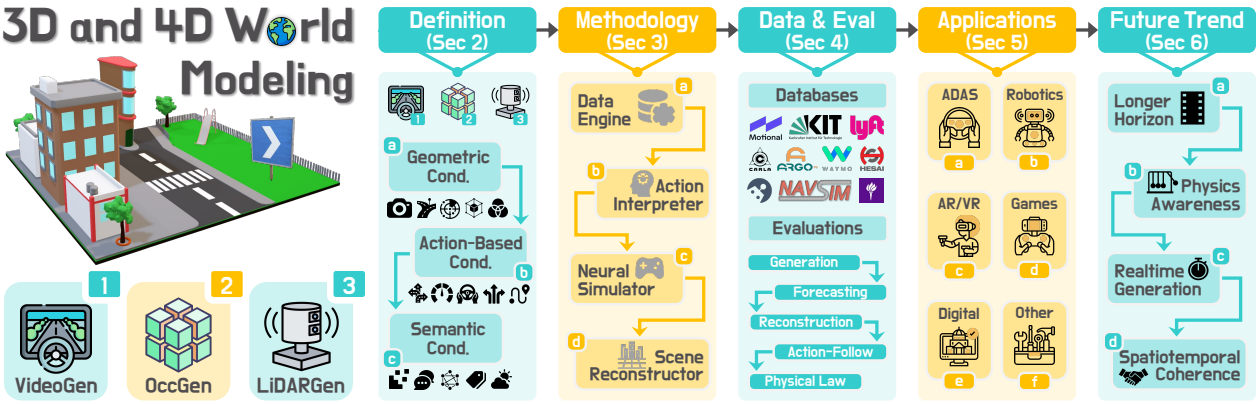


Figure 1 Outline of the survey. This work focuses on **native 3D and 4D representations**: video streams, occupancy grids, and LiDAR point clouds, guided by geometric ($\mathcal{C}_{\text{geo}}$), action-based ($\mathcal{C}_{\text{act}}$), and semantic ($\mathcal{C}_{\text{sem}}$) conditions (Sec. 2). Methods are framed under two paradigms, *generative* (synthesis from observations and conditions) and *predictive* (forecasting from history and actions), and grouped into four functional types (Sec. 3). We cover three modality tracks and standardize evaluations (Sec. 4), practical applications (Sec. 5), and future trends (Sec. 6) across diverse generation, forecasting, and downstream task perspectives.

1 Introduction

World modeling has emerged as a fundamental task in AI and robotics, aiming towards the ability to understand, represent, and anticipate the dynamic environments they inhabit [17, 30, 290]. Recent advancements in generative modeling techniques, including VAEs, GANs, diffusion models, and autoregressive models, have significantly enriched the field by enabling sophisticated generation and prediction capabilities [72, 284].

Much of this progress, however, has been centered on 2D data, primarily images or videos [60, 250, 286]. Real-world scenarios, in contrast, are inherently in 3D space and dynamic, often requiring models that leverage **native 3D and 4D representations**. These include RGB-D imagery [32, 102, 321], occupancy grids [36, 258, 339], and LiDAR point clouds [21, 91, 176], as well as their sequential forms that capture temporal dynamics [24, 451]. These modalities offer explicit geometry and physical grounding, which are indispensable for embodied and safety-critical systems such as autonomous driving and robotics [95, 97, 175, 198, 229, 481, 484].

Beyond these native formats, world modeling has also been explored in adjacent domains [92, 93, 438]. Some works address video, panoramic, or mesh-based data, with systems of this kind providing large-scale, general-purpose video-mesh generation capabilities. In parallel, another line of research focuses on 3D object generation for asset creation, which specializes in controllable and high-fidelity object synthesis. Meanwhile, industrial projects from leading companies have launched ambitious world modeling initiatives that target practical applications ranging from interactive robotics and immersive simulation to large-scale digital twins [1, 5, 8, 12, 15, 302], underscoring the growing importance of this field in both academia and industry.

Despite this momentum, the term "world model" itself remains ambiguous, with inconsistent usage across the literature [72, 92, 399]. Some works narrowly interpret it as generative models for sensory data (*e.g.*, images and videos), while others broaden the scope to include predictive forecasting, simulators, and decision-making frameworks [111, 163, 234, 342, 413]. Moreover, existing surveys largely emphasize 2D or vision-only modalities [60, 494], leaving the unique challenges and opportunities of native 3D and 4D data underexplored. This has led to a fragmented body of literature lacking a unified framework or taxonomy.

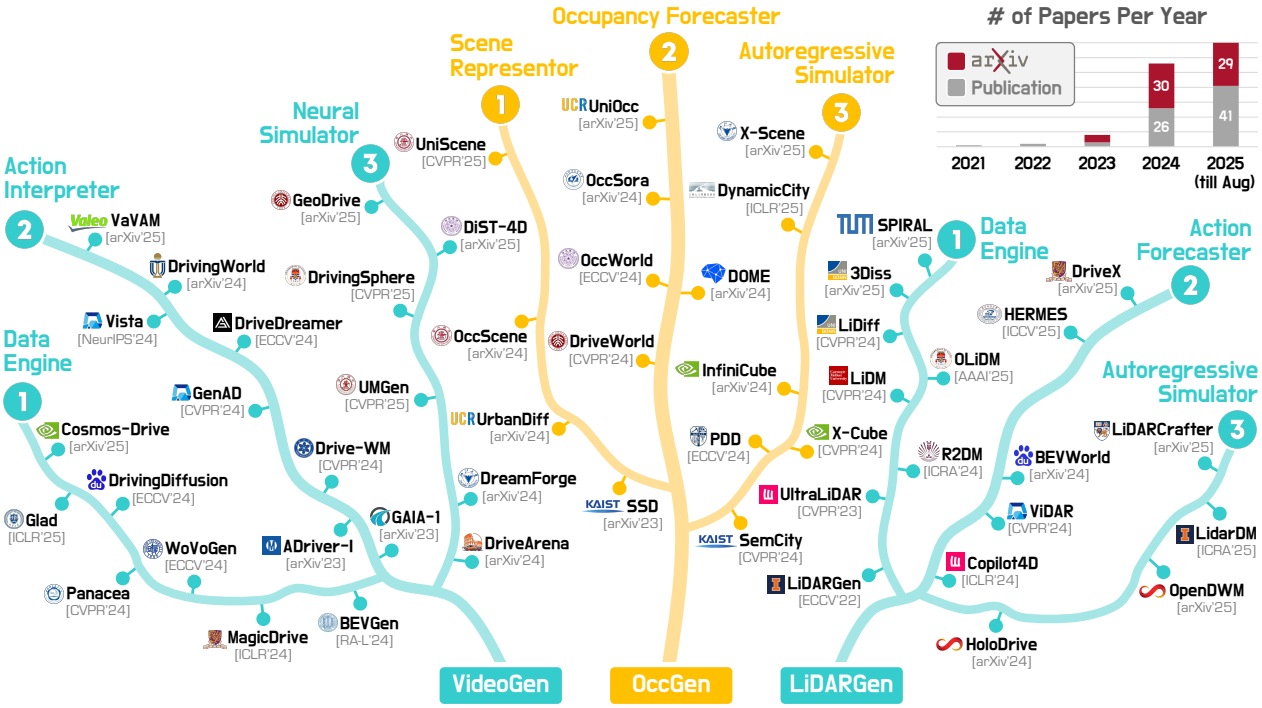


Figure 2 Summary of representative video-based generation **(VideoGen)**, occupancy-based generation **(OccGen)**, and LiDAR-based generation **(LiDARGen)** models from existing literature. For the complete list of related methods and discussions on their specifications, configurations, and technical details, kindly refer to Sec. 3.1, Sec. 3.2, and Sec. 3.3.

Complementary surveys chart adjacent territory, including the foundations and laws of agentic world modeling [62], broad treatments of world-model architectures, methodologies, and reasoning paradigms [496], the frontiers of interactive *video* world modeling [217], and vision-language-action models for autonomous driving [133], which our focus on native 3D and 4D representations is designed to complement.

Why do native 3D and 4D matter?

Unlike 2D projections, native 3D/4D signals encode metric geometry, visibility, and motion in coordinates where physics acts [24, 205]. This makes them *first-class carriers* of constraints needed for actionable modeling: multi-view and egocentric consistency, rigid-body and non-rigid kinematics, scene-scale occlusion reasoning, and map/topology adherence. In safety-critical settings, agents must not only produce photorealistic frames but also obey geometry, causality, and controllability; RGB-D, occupancy, and LiDAR provide the inductive bias to satisfy these requirements. Sec. 2 will formalize these representations and the conditioning signals $(\mathcal{C}_{\text{geo}}, \mathcal{C}_{\text{act}}, \mathcal{C}_{\text{sem}})$ we use throughout the survey.

Position in the broader landscape.

The adjacent lines – video/panorama/mesh world models [141] and object-centric 3D asset generators [422] – are complementary: they supply appearance, topology, and assets, while native 3D/4D world models supply geometry-grounded dynamics and interaction [451, 463]. Practical systems increasingly compose these capabilities: mesh/panorama worlds initialized from assets, then *driven* by occupancy- or LiDAR-based dynamics, or video models constrained by 3D priors for view and motion correctness. Our scope centers on the latter – native 3D/4D – while acknowledging and cross-referencing where cross-pollination occurs.

From conditions to functions.

A common pain point in the field is conflating *"what the model consumes"* (conditions) with *"what the model does"* (function). We therefore separate the roles of geometry/action/semantics conditions (Table 1) from functional types. Sec. 3 organizes methods by representation modality –**VideoGen**, **OccGen**, **LiDARGen** – and then by **four** functional roles: [1]*Data Engines* (diverse scene synthesis under $\mathcal{C}_{\text{geo}}, \mathcal{C}_{\text{sem}}, \mathcal{C}_{\text{act}}$), [2]*Action Interpreters* (forecasting under $\mathcal{C}_{\text{act}}$ with history), [3]*Neural Simulators* (closed-loop rollouts with policy-in-the-loop), and [4]*Scene Reconstructors* (completion/retargeting from partial 3D/4D observations). This decoupling lets us compare heterogeneous methods on common axes of fidelity, consistency, controllability, and scalability.

Contributions. To address the aforementioned gaps, this survey presents the first comprehensive review specifically dedicated to **3D and 4D world modeling and generation**. The primary contributions of this survey are **threefold**:

- We establish precise definitions for "world models" and "3D/4D world modeling", providing the research community with consistent terminology and conceptual clarity.
- We propose a hierarchical taxonomy of methodologies, categorizing current approaches based on their representation modalities – namely, world modeling based on VideoGen, OccGen, and LiDARGen models.
- We provide extensive coverage of datasets and evaluation protocols specifically tailored for 3D and 4D scenarios, enabling a thorough benchmarking of existing and future world modeling and generation approaches.

Scope. Distinct from previous surveys, which predominantly focus on 2D generative models [211, 366, 494] or broadly define world modeling within limited contexts [88, 374, 395, 461, 462], this survey explicitly targets methodologies that utilize native 3D and 4D representations. This specialized focus includes approaches leveraging RGB-D, volumetric occupancy grids, LiDAR point clouds, and their spatiotemporal forms. By highlighting these modalities, our survey not only fills a critical knowledge gap but also serves as a foundational reference for researchers aiming to develop robust and generalizable 3D/4D generative models.

Organization. The remainder of this survey is organized as follows. Sec. 2 provides preliminaries, detailing fundamental concepts, definitions, and key generative paradigms relevant to world modeling. Sec. 3 introduces a new and hierarchical taxonomy, detailing VideoGen, OccGen, and LiDARGen methodologies, providing comparative analyses and insights into their respective strengths and limitations. Sec. 4 systematically summarizes and categorizes widely used datasets and evaluation metrics critical for world modeling tasks, as

well as benchmarking recent methods in this related area. Sec. 5 reviews practical applications of 3D and 4D world models across autonomous driving, robotics, and simulation environments. Sec. 6 discusses major challenges and highlights promising future research directions, paving the way for continued innovation in the field. Finally, Sec. 7 concludes the key discussions drawn in this survey.

2 Preliminaries

In this section, we define critical concepts and establish unified mathematical notations essential for understanding 3D and 4D world modeling. This includes detailed descriptions of the key representations, definitions of generative and predictive world models, and model categorizations.

2.1 3D and 4D Representations

To systematically analyze 3D/4D world models, we first introduce the fundamental scene representations that serve as inputs, outputs, or intermediate states in generation and prediction. These representations differ in how they capture spatial geometry, temporal dynamics, and semantic context.

Video Streams. A video is denoted as $\mathbf{x}_v \in \mathbb{R}^{T\times H\times W\times C}$, where T is the number of frames, and H, W, C are the frame height, width, and channels. Unlike conventional 2D videos, 3D/4D modeling emphasizes *geometric coherence* and *temporal consistency* to ensure physically plausible simulations and accurate forecasting [127, 365, 449].

Occupancy Grids. A static occupancy grid is represented as $\mathbf{x}_o \in \{0,1\}^{X\times Y\times Z}$, where each voxel indicates whether a location is occupied [258, 289]. Sequential occupancy grids $\mathbf{x}_o^t \in \{0,1\}^{T\times X\times Y\times Z}$ extend this into 4D, capturing scene evolution over time. Such voxelized geometry enforces spatial constraints, making them well-suited for physics-consistent scene generation. We emphasize that occupancy plays *two distinct roles* in this survey: here it is a primary generation or forecasting *target* (a full-resolution input/output representation), whereas in Sec. 2.2 an occupancy volume may instead serve as a geometric *conditioning* signal $\mathcal{C}_{\text{geo}}$ that constrains the synthesis of another target modality rather than being the output itself.

LiDAR Point Clouds. A LiDAR-acquired scan is expressed as $\mathbf{x}_l = \{(x_i, y_i, z_i)\}_{i=1}^{N}$, where (x_i, y_i, z_i) are the Cartesian coordinates in 3D space [295]. Sequential LiDAR $\mathbf{x}_l^t = \{(x_i, y_i, z_i, t_i)\}_{i=1}^{N_t}$ further records the timestamp t_i, enabling precise modeling of motion and interactions [167, 406]. Unlike RGB images, LiDAR captures geometry directly and remains robust to texture, lighting, or weather variations [174, 198].

Neural Representations. Implicit scene encodings, such as neural radiance fields (NeRF) [259] and Gaussian splatting (GS) [165], model continuous volumetric fields or explicit Gaussian primitives. NeRF maps a ray origin $\mathbf{r}$ and direction $\mathbf{d}$ to color $\mathbf{c}$ and density σ, while GS represents the scene as a set of Gaussians parameterized by position, covariance, and color. Temporal extensions add dynamic components, enabling realistic 4D reconstructions and simulations.

2.2 Definition of World Modeling in 3D and 4D

A recurring source of ambiguity in the literature is the conflation of *3D/4D generation*, *3D/4D world modeling*, *scene simulation*, and *world models*. We draw explicit boundaries among them. *3D/4D generation* synthesizes spatial or spatiotemporal signals (pixels, voxels, or points) and is judged primarily on fidelity and diversity, without necessarily modeling dynamics or supporting interaction. *3D/4D world modeling* additionally captures how a scene *evolves*, predicting future states from past observations and actions; crucially, this prediction need not be carried out in observation space. *Scene simulation* refers to the closed-loop, interactive rollout of an environment driven by agent actions, where generated states are fed back as inputs to subsequent steps. *World models*, in the broadest sense, denote the umbrella notion of an internal, action-conditioned predictive model of environment dynamics. Under this view, a generator that merely synthesizes pixels or voxels is not, by itself, a world model: observation-space reconstruction is *sufficient but not necessary*. Non-reconstructive paradigms, most notably Joint-Embedding Predictive Architectures (JEPA) [7, 8, 17], instead predict in a learned representation space and forgo pixel/voxel-level reconstruction. In this survey, we focus on *native 3D/4D* world models that operate over explicit geometric representations (video, occupancy, and LiDAR),

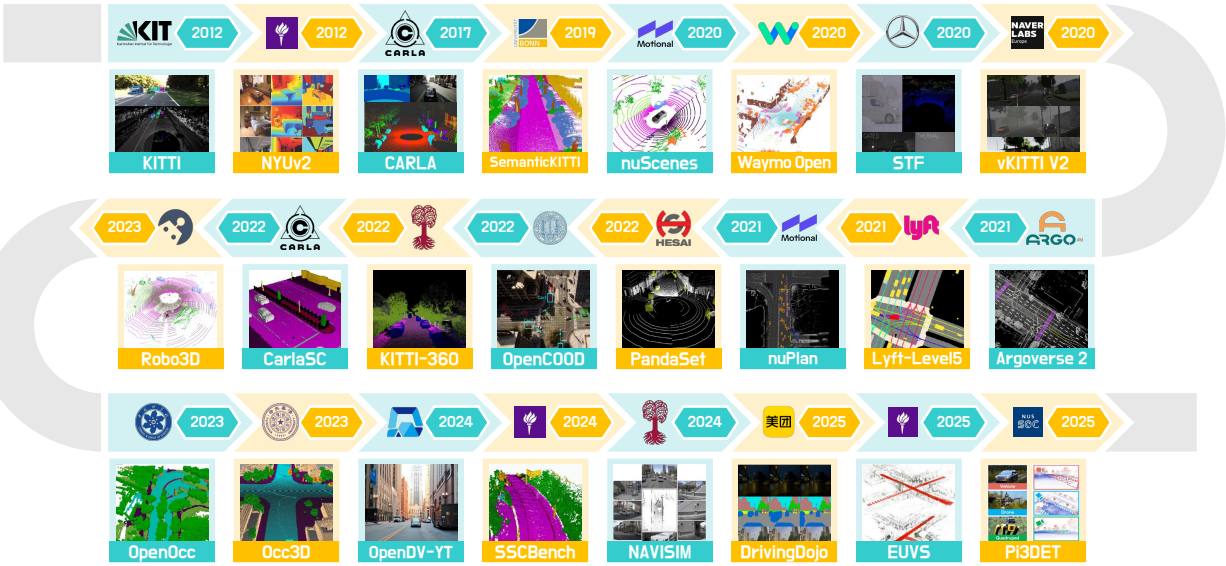


Figure 3 Summary of existing **datasets & benchmarks** used for training and evaluating **VideoGen**, **OccGen**, and **LiDARGen** models. For detailed configurations and statistics, kindly refer to Table 5. Images adopted from the original papers.

while explicitly recognizing the representation-space paradigm as a complementary and increasingly important direction.

The above scene representations form the structural backbone of 3D/4D world models. In practice, generating or forecasting them requires additional *conditions* – auxiliary signals that constrain spatial structure, describe agent behavior, or define high-level semantics. As summarized in Table 1, these conditions are typically grouped into:

- **geometric** $\mathcal{C}_{\text{geo}}$**:** specifying spatial layout such as camera pose, depth maps, or occupancy volumes (used here as a conditioning prior, as distinct from occupancy as a generation target in Sec. 2);
- **action-based** $\mathcal{C}_{\text{act}}$**:** describing ego-vehicle or agent motion via trajectories, control commands, or navigation goals;
- **semantic** $\mathcal{C}_{\text{sem}}$**:** providing abstract scene intent such as textual prompts, scene graphs, or environment attributes.

These signals can be used independently or in combination, shaping the realism, controllability, and diversity of the generated or forecasted scenes in 3D and 4D.

2.2.1 Model Definitions

Depending on the modeling objective, 3D/4D world models generally fall into two complementary paradigms:

Generative World Models focus on synthesizing plausible scenes from scratch or from partial observations, guided by multimodal conditions. This process can be formulated as:

$$\mathcal{G}(\mathbf{x}_i, \mathcal{C}_{\text{geo}}, \mathcal{C}_{\text{act}}, \mathcal{C}_{\text{sem}}) \rightarrow \mathcal{S}_g, \tag{1}$$

where $\mathbf{x}_i$ denotes the optional input representation, with $i \in \{\varnothing, v, o, l\}$, *e.g.*, noise, partial video, occupancy, or LiDAR data. $\mathcal{C}_{\text{geo}}$, $\mathcal{C}_{\text{act}}$, and $\mathcal{C}_{\text{sem}}$ correspond to the geometric, action, and semantic conditions. The output $\mathcal{S}_g$ is a generated 3D/4D scene, such as a video sequence, occupancy grid, or LiDAR sweep sequence.

Predictive World Models instead aim to forecast the future evolution of the scene based on historical observations, often under action conditions that describe planned or executed agent behavior. This process is formulated as:

$$\mathcal{P}(\mathbf{x}_i^{-t:0}, \mathcal{C}_{\text{act}}) \rightarrow \mathcal{S}_p^{1:k}, \tag{2}$$

Table 1 Summary of the rich collection of conditions used by **VideoGen**, **OccGen**, and **LiDARGen** models. The conditions are categorized into **three** main groups: geometric conditions, action-based conditions, and semantic conditions. The tasks are ❶ video generation (Sec. 3.1), ❷ occupancy generation (Sec. 3.2), and ❸ LiDAR generation (Sec. 3.3).

Group	Condition	Definition	Task
Geometry ($\mathcal{C}_{\text{geo}}$)	C Camera Pose	Position and orientation of the camera in world coordinates, controlling viewpoint	❶
	D Depth Map	Per-pixel depth values providing scene geometry constraints	❶
	B BEV Map	Bird's-eye-view geometric representation of the scene	❶ ❷ ❸
	H HD Map	High-resolution semantic map with detailed road layout and traffic elements	❶ ❸
	3 3D Bounding Box	Object bounding boxes in 3D, defining positions, sizes, and orientations of objects	❶ ❸
	F Flow Field	Optical or scene flow encoding per-pixel or per-point motion between frames	❶ ❷
	P Past Occupancy	Historical occupancy grids or voxel maps capturing prior scene geometry	❷ ❸
	L LiDAR Pattern	Sensor scan configuration including beam count, FOV, and resolution	❸
	O Object Coordinate	Set of Cartesian coordinates of instances from LiDAR point clouds	❶ ❸
	P Partial Point Cloud	Incomplete LiDAR point set capturing only a subset of the full 3D scene geometry	❸
	R RGB Frame	Single color image frame from a monocular or multi-camera setup	❶ ❷ ❸
	S Surface Mesh	Triangular mesh or equivalent explicit geometry representation of the scene	❶ ❷ ❸
Action ($\mathcal{C}_{\text{act}}$)	T Ego-Trajectory	The planned or recorded path of the ego vehicle over time	❶ ❷ ❸
	V Ego-Velocity	The speed and direction of the ego movement	❶ ❷
	A Ego-Acceleration	Rate of change of ego velocity, describing linear acceleration or deceleration	❶
	S Ego-Steering	The steering angle or input controlling the ego direction	❶
	C Ego-Command	The control instructions given to the ego vehicle	❶ ❷ ❸
	R Route Plan	High-level navigation path through the environment, often from a planner	❶
	A Action Token	Encoded discrete actions or instructions influencing scene evolution	❶
	S Scan Path	Predefined movement or sweep pattern during LiDAR acquisition	❸
Semantics ($\mathcal{C}_{\text{sem}}$)	S Semantic Mask	Pixel-/occupancy-/point-wise semantic categories	❶ ❷ ❸
	T Text Prompt	Natural language input specifying scene attributes, objects, or actions	❶ ❷ ❸
	G Scene Graph	Graph representation of scene entities and their spatial/semantic relationships	❷ ❸
	O Object Label	Class category annotation assigned to an object instance in the scene	❶ ❷ ❸
	W Weather Tag	Discrete label describing environmental conditions such as sunny, rainy, or foggy	❶ ❷
	M Material Tag	Classification of surface materials influencing appearance or LiDAR reflectance	❶ ❸

where $\mathbf{x}_i^{-t:0}$ represents observations from the past t steps to the current step, and $\mathcal{C}_{\text{act}}$ encodes agent actions (*e.g.*, control commands or planned trajectories). The model outputs $\mathcal{S}_p^{1:k}$, the predicted scene representations over k future steps.

Together, these two paradigms capture the dual capability of world models: the ability to *imagine* diverse and controllable worlds (generative), and to *anticipate* their plausible future evolution under specific conditions (predictive).

2.2.2 Model Categorizations

Building on the generative and predictive paradigms, existing approaches can be further divided into **four functional types**. They differ in how they utilize historical observations, the nature of conditioning signals ($\mathcal{C}_{\text{geo}}, \mathcal{C}_{\text{act}}, \mathcal{C}_{\text{sem}}$), and whether they operate in an open-loop or closed-loop setting.

Type 1: Data Engines

Generate diverse 3D/4D scenes from geometric and semantic cues, optionally with action conditions.
- **Inputs:** $\mathcal{C}_{\text{geo}}$ (geometric cond.), $\mathcal{C}_{\text{act}}$ (action cond., optional), and $\mathcal{C}_{\text{sem}}$ (semantic cond.)
- **Output:** $\mathcal{S}_g$ (generated scene)

Focus on *plausibility* and *diversity* for large-scale data augmentation and scenario creation.

Type 2: Action Interpreters

Forecast future 3D/4D world states from historical observations under given action conditions.
- **Inputs:** $\mathbf{x}_i^{-t:0}$ (historical observations) and $\mathcal{C}_{\text{act}}$ (action cond.)
- **Output:** $\mathcal{S}_p^{1:k}$ (predicted sequence)

Enable *action-aware forecasting* for trajectory planning, behavior prediction, and policy evaluation.

Type 3: Neural Simulators

Iteratively simulate closed-loop agent-environment interactions by generating successive scene states.
- **Inputs:** $\mathcal{S}_g^t$ (current scene state) and π_{agent} (agent policy)
- **Output:** $\mathcal{S}_g^{t+1}$ (next scene state)

Support *interactive simulation* for autonomous driving, robotics, and immersive XR training.

Type 4: Scene Reconstructors

Recover complete and coherent 3D/4D scenes from partial, sparse, or corrupted observations.
- **Inputs:** $\mathbf{x}_i^p$ (partial observations) and $\mathcal{C}_{\text{geo}}$ (optional geometric cond.)
- **Output:** $\hat{\mathcal{S}}_g$ (completed scene)

Facilitate *interactive tasks* on high-fidelity mapping, digital twin restoration, and post-event analysis.

Together, these four categories outline the functional landscape of 3D/4D world modeling. While all aim to produce physically and semantically coherent scenes, they differ in how they leverage past observations, conditioning signals, and interaction loops – serving applications ranging from large-scale data synthesis and policy evaluation to interactive simulation and scene restoration.

2.3 Generative Models

Generative models form the algorithmic core of 3D/4D world modeling, enabling agents to *learn*, *imagine*, and *forecast* sensory data under diverse conditions. They provide the mechanisms to synthesize realistic and physically plausible scenes, with different paradigms offering distinct trade-offs in quality, controllability, and efficiency. Representative families include variational autoencoders, generative adversarial networks, diffusion models, and autoregressive models.

Variational Autoencoders (VAEs) [171] learn a structured latent space via probabilistic encoding and decoding. Given input $\mathbf{x}$, the encoder defines a variational posterior $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mu_\phi(\mathbf{x}), \text{diag}(\sigma_\phi^2(\mathbf{x})))$ and samples $\mathbf{z}$ using the reparameterization trick: $\mathbf{z} = \mu_\phi(\mathbf{x}) + \sigma_\phi(\mathbf{x}) \odot \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, I)$. The decoder $p_\theta(\mathbf{x}|\mathbf{z})$ reconstructs the input, and the model is trained to maximize the variational lower bound that balances reconstruction fidelity and latent regularization:

$$\log p_\theta(\mathbf{x}) \geq \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z})). \tag{3}$$

VAEs offer stable training and interpretable latent spaces, but may produce blurrier samples compared to other paradigms.

Generative Adversarial Networks (GANs) [105] generate data via a min–max game between a generator G_θ and discriminator D_ϕ. The generator maps latent variables $\mathbf{z} \sim p(\mathbf{z})$ to the data space, aiming to fool D_ϕ, while the discriminator distinguishes real from synthetic samples:

$$\min_G \max_D \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))]. \tag{4}$$

GANs can produce high-fidelity result samples but often suffer from training instability and mode collapse issues.

Diffusion Models (DMs) [123, 322] learn to reverse a gradual noising process. The forward process corrupts $\mathbf{x}_0$ into $\{\mathbf{x}_1, \ldots, \mathbf{x}_T\}$ via $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$, where β_t follows a variance schedule. The reverse process $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is trained to denoise, minimizing:

$$\mathbb{E}_{\mathbf{x},\boldsymbol{\epsilon},t}[\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2]. \tag{5}$$

DMs provide strong stability and sample quality, though inference can be slow due to iterative sampling.

Autoregressive Models (ARs) [338, 346] factorize the joint distribution as $p(\mathbf{x}) = \prod_{i=1}^n p(x_i \mid x_{<i})$, predicting each element conditioned on all previous ones. Transformer-based ARs offer exact likelihood estimation and

flexible sequence modeling, but suffer from slow generation since samples are produced sequentially. Recent advances have adapted ARs to spatial and temporal tokens, making them well-suited for structured 3D scene generation and forecasting.

Adaptation to native 3D/4D. Realizing these paradigms over high-dimensional 3D/4D data hinges on representation-specific design. VAEs most often act as *latent tokenizers* – video, triplane, and voxel autoencoders that compress volumes into compact codes so that downstream diffusion or autoregressive priors become tractable over occupancy, LiDAR, and long video. GANs, with single-step sampling, remain attractive for efficiency-critical synthesis (*e.g.*, fast LiDAR range-image and BEV/occupancy generation) and as auxiliary sharpening objectives, though diffusion has largely superseded them for high-fidelity generation. Diffusion models dominate native 3D/4D generation through 3D- and video-aware UNets and diffusion transformers, with latent, few-step, and flow-matching variants that ease sampling cost while enabling controllable synthesis of multi-view video, 4D occupancy, and point clouds. Autoregressive models, via spatial and temporal tokenizations (*e.g.*, serialized occupancy/LiDAR tokens and next-scale prediction), suit structured, long-horizon forecasting and the causal dependencies of closed-loop rollouts, and integrate naturally with language interfaces for instruction-conditioned generation.

Summary. These paradigms form the algorithmic backbone for world models. Their differences in structure, training stability, and inference efficiency directly shape how 3D environments can be synthesized, forecasted, and controlled. As we move into native 3D/4D domains, these trade-offs are magnified, since scalability, controllability, and multi-modal integration are critical to constructing reliable world models for embodied AI and simulation.

3 Methods: A Hierarchical Taxonomy

In this section, we standardize and categorize existing 3D and 4D world modeling approaches based on their representation modalities. This includes descriptions and discussions of world modeling based on **Video Generation** (Sec. 3.1), **Occupancy Generation** (Sec. 3.2), and **LiDAR Generation** (Sec. 3.3) models, respectively.

3.1 World Modeling from Video Generation

Video-based generation has emerged as a new paradigm, offering visual cues and temporal dynamics to model complex real-world scenarios. By generating multi-view or egocentric video sequences, these models can synthesize training data, predict future outcomes, and create interactive simulation environments. Based on their primary function, existing methods can be grouped into **three** categories: ❶ **Data Engines**, ❷ **Action Interpreters**, and ❸ **Neural Simulators**. Table 2 summarizes existing models under these domains.

3.1.1 Data Engines

Generative 3D data engines focus on generating diverse and controllable driving scenes to support perception, planning, and simulation [73, 96, 97, 154, 181, 190, 330, 420]. Research in this direction covers three major applications.

Perception Data Augmentation. Generative scene synthesis alleviates real-world data scarcity and addresses long-tail perception challenges. Early work focused on BEV-guided realistic street scenes. BEVGen [330] uses an autoregressive transformer and cross-view transformation to produce spatially consistent surrounding images aligned with a given BEV layout. BEVControl [420] centers on diffusion models to boost the quality of synthetic data, particularly for augmenting challenging long-tail scenarios. Subsequently, MagicDrive [97] made significant progress in driving scene generation and data augmentation, combining 3D geometry and semantic descriptions, and camera poses to generate high-fidelity images. Later work introduced finer conditioning. For instance, SyntheOcc [190] uses 3D semantic multi-plane images for comprehensive, spatially aligned conditioning, and PerLDiff [447] proposes perspective-layout diffusion models that fully leverage perspective 3D geometry to enhance realism and consistency. On the other hand, approaches such as Panacea [375], DrivingDiffusion [195], and SubjectDrive [136] introduce 4D attention, keyframes, and subject control to improve the temporal consistency and data diversity of 3D controllable multi-view videos. MyGo [434] injects camera-pose conditioning to yield consistent and controllable multi-view driving videos, and

NoiseController [73] proposes multi-level noise decomposition and multi-frame collaborative denoising to enhance spatiotemporal coherence.

For long-horizon video generation, DiVE [154], MagicDrive-V2 [96], and Cosmos-Drive [302] leverage the flexibility and scalability of DiT to produce longer videos. Glad [393] uses latent-variable propagation, and STAGE [353] uses hierarchical temporal feature transfer to generate long videos in a streaming fashion. Others like UniScene [181] and BEVWorld [457] explore multi-modal data synthesis to broaden applications, supporting downstream perception tasks that leverage information from multiple modalities. AutoScape [46] enforces geometric consistency to produce long-horizon driving scenes, while OmniDrive [256] generates multi-view driving videos via unified latent co-compression. Complementing these generators, recent studies reframe the driving world model explicitly as a synthetic data generator and revisit how its rollouts should be curated to benefit downstream perception [442]. These advances enable robust, scalable autonomous driving perception systems by delivering diverse, controllable, and long-horizon training data that capture real-world variability. Pushing this further, OpenLong-Tail [220] generatively scales scarce long-tail driving data by lifting heterogeneous single- and partial-view clips into multi-view training samples. Orthogonal to data curation, MiniWorld [467] instead targets training accessibility, democratizing how video world models themselves are trained from scratch rather than only how their training data is sourced.

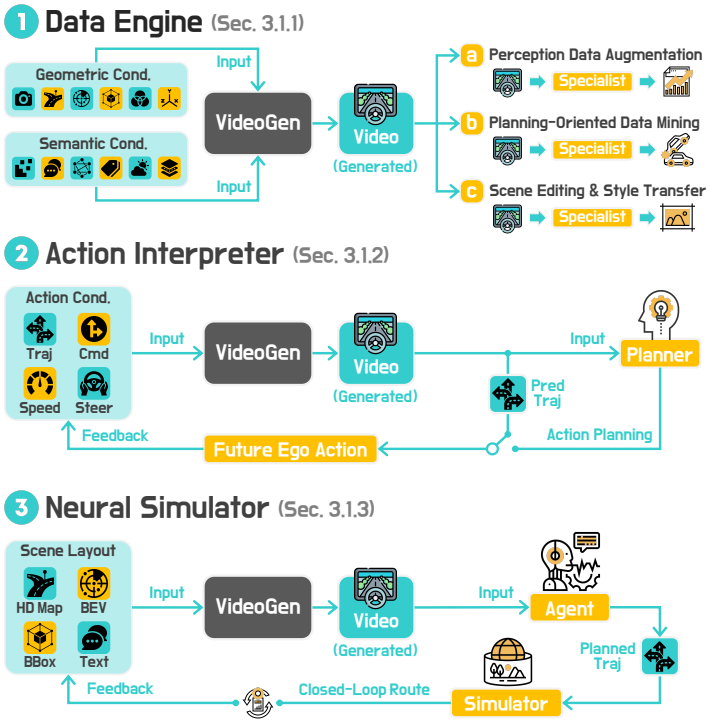


Figure 4 The categorization of **VideoGen** models based on functionalities, including data engines (Sec. 3.1.1), action interpreters (Sec. 3.1.2), and neural simulators (Sec. 3.1.3).

Planning-Oriented Data Mining. Beyond perception, data engines also mine rare and safety-critical scenarios for planning. Delphi [245] employed a diffusion-based long video generation framework and a failure-case-driven approach utilizing pre-trained visual language models to synthesize data similar to failure scenarios, thereby enhancing sample efficiency and planning performance for end-to-end autonomous driving systems. DriveDreamer-2 [464] converted user queries into agent trajectories via a large language model, which are then used to produce traffic-compliant HDMaps for corner case generation. Nexus [489] simulated both regular and challenging scenarios from fine-grained tokens with independent noise states to improve reactivity and goal conditioning and collected a specialized corner-case dataset to complement challenging scenario generation. Challenger [408] exploited a physics-aware multi-round trajectory refinement to identify adversarial maneuvers and a tailored scoring function to promote realistic yet challenging behaviors compatible with downstream video synthesis. OmniSCS [75] instead casts safety-critical scenario synthesis as editing within a fully controllable driving world, enabling omni-directional generation of rare, hazardous cases.

Scene Editing & Style Transfer. Many existing methods [239, 375, 488] also take world models for scene editing and style transfer to enrich the toolkit for autonomous driving simulation and data augmentation. Early methods primarily utilized scene descriptions [97] or reference images [48] for basic appearance modifications (*e.g.*, weather, lighting) and relied on bounding boxes or HD maps [420] for element-level adjustments. However, newer approaches explore richer representations for precise scene manipulation and diverse appearance control. WoVoGen [239] ensures cross-sensor consistency through world volume-aware synthesis, while SyntheOcc [190] employs occupancy grids for occlusion-aware scene editing. SimGen [488] bridges sim-to-real gaps via simulator-conditioned cascade diffusion, and DrivePhysica [432] simulates complex driving scenarios (*e.g.*, cut-ins) using CARLA and introduces motion representation learning and instance flow guidance for temporal consistency.

Complementing these, GeoDrive [43] integrates explicit 3D geometry conditions and dynamic editing to enable interactive trajectory and object manipulation. More recent editors target object-level insertion and real-world grounding: DriveWeaver [156] performs point-conditioned video inpainting for controllable vehicle insertion, while RealityBridge [387] couples editable 3D Gaussian Splatting simulations with real-world videos for photorealistic, editable rollouts. Weather remains a persistent editing target: RealWeather [275] translates driving footage across weather conditions while preserving scene fidelity, and muSync-GS [53] synchronizes physics with video synthesis to render safety-critical weather and road-geometry hazards that are otherwise costly to capture.

Beyond Driving: General-Purpose and Embodied Data Engines. Data engines are also expanding past the driving domain. ABot-3DWorld [327] turns text, image, or video into explorable 3D worlds through a unified spatial generative primitive, and Sekai2 [119] pairs long, camera-trajectory-annotated video corpora with training recipes for long-horizon, camera-controllable world exploration. Xiaomi-Robotics-U0 [196] adapts foundation video generators to embodied scenarios by enforcing multi-view consistency, geometric coherence, and robot-embodiment constraints that natural-scene video generators do not respect. EmbodiedVAE [244] targets the same gap at the representation level, disentangling a video VAE so that its latent space captures the structure embodied manipulation actually needs. H2R-Bench [306] extends the data-engine role to cross-embodiment transfer, generating robot-consistent manipulation video directly from abundant egocentric human footage.

Explicit Geometry as a Conditioning Substrate. A distinct recent trend replaces the weak conditioning signals of earlier generators with an explicit geometric proxy that the generative model then refines. SpatialCrafter [86] argues that sparse point clouds or 2D panoramas underdetermine the target scene, and instead predicts a spatially aligned global 3D proxy before re-framing a video diffusion model as a deferred refiner that adds high-frequency appearance on top of proxy-defined geometry; the accompanying 115K-scene corpus is assembled specifically for explorable image-to-scene generation. 4DStreamCtrl [192] pushes the control interface in the same direction, exposing camera and object motion jointly as online 4D control rather than as separate camera-parameter or object-trajectory conditions. NeoWorld-Pro [120] takes the programmatic route, turning monocular images into interactive scenes for embodied simulation. Read together with the geometry-conditioned editors above, these systems indicate that the field is converging on explicit 3D or 4D structure as the conditioning substrate for scene synthesis, with the generative model responsible for appearance rather than for inferring geometry implicitly.

4Director [38] makes the geometric control interface more explicit: it reconstructs each marked object once as a canonical mesh, applies prescribed rigid transformations, and renders depth sequences to condition a video generator. Its Motion Adapter supplies appearance and non-rigid dynamics that the rigid scaffold omits. This separation improves the interpretability of camera and object control, while also delimiting the claim: following a prescribed 3D trajectory does not by itself establish that the generator predicts physically valid object dynamics.

Sensor-Specific and Long-Horizon Data Engines. Two further constraints shape what a data engine must deliver. The first is sensor geometry: surround coverage is increasingly obtained from a small number of fisheye cameras, yet generative models are trained almost exclusively on rectilinear imagery. MIVIFI [272] formalizes multi-view fisheye image generation and bridges the perspective and fisheye domains, so that rare configurations can be synthesized without the cost of full 3D simulation. The second is horizon: ReWorld [56] places eight heterogeneous sources, from Unreal fly-throughs to real footage, on a single physical action scale so that a given control input moves the camera the same distance in every source, and uses palindrome trajectories to supply the revisit evidence that long-horizon memory training requires. Both illustrate that the binding constraint on data engines is shifting from raw visual quality toward the metric and sensor-level consistency that downstream training actually consumes.

3.1.2 Action Interpreters

Action-driven generation models bridge agent intentions and environmental dynamics through action-guided world generation and forecast-driven action planning, enabling outcome anticipation and unifying low-level maneuvers and reasoning by mapping controls to plausible futures.

Action-Guided Video Generation. Action-conditioned generation models empower agents to predict future outcomes based on intended maneuvers, effectively bridging low-level control inputs with high-fidelity video

rollouts of plausible futures. GAIA-1 [127] pioneered a generative model that fuses video, text, and action inputs to synthesize realistic driving scenarios with detailed control over ego-vehicle behavior and scene attributes. GAIA-2 [307] expanded this framework to include agent configurations, environmental factors, and road semantics. GenAD [418] further enhanced generalization by releasing the OpenDV dataset alongside a predictive model that supports zero-shot, language- and action-conditioned predictions. Vista [98] applies robust action conditioning across diverse scenarios, while GEM [118] delivers multimodal outputs with precise ego-motion control, and MaskGWM [274] boosts fidelity and long-horizon predictions using mask-based diffusion.

To address error accumulation in long video synthesis, InfinityDrive [114] and Epona [449] proposed memory injection and a chain-of-forward training strategy, respectively. Cycle-World [325] attacks the same problem from the training-objective side, imposing a reverse-prediction cycle-consistency loss so the model is penalized for information a forward rollout would otherwise silently discard. In addition, DrivingWorld [134] generates scenarios from predefined trajectories, functioning as a neural driving simulator. Other approaches, such as DriVerse [194], MiLA [351], PosePilot [160], and LongDWM [363], focus on trajectory alignment, temporal stability, pose controllability, and depth-free guidance. OmniNWM [183] pushes toward an omniscient navigation world model that jointly generates RGB, semantics, depth, and occupancy under action control, while GenieDrive [431] leverages 4D occupancy as a physics-aware guide for video generation. Beyond conditioning on the ego trajectory alone, recent work argues that surrounding-agent motions are equally indispensable and unifies ego and other-vehicle trajectories within a shared video latent space for coherent multi-agent rollouts [492]. Collectively, these advances drive action-conditioned generation toward better precision, temporal coherence, and robustness. Deployment cost is a growing concern alongside quality: Adaptive-WAM [6] exits early from the video-diffusion backbone once intermediate features are informative enough for the downstream ego trajectory, ForgeWM [197] trains a few-step causal generator so action-conditioned rollouts no longer require the full iterative denoising schedule, and DriveCache [417] caches intermediate activations across timesteps to accelerate repeated inference without retraining.

CoDrive [255] extends trajectory-conditioned generation from several cameras on one vehicle to observations shared by multiple vehicles. It combines within-vehicle and cross-vehicle attention and expresses camera trajectories in a common world coordinate system. Its CoDrive-Bench consequently evaluates cross-agent geometry and instance consistency alongside trajectory control, exposing a constraint that independent ego-view rollouts cannot enforce.

Beyond Driving: Action-Conditioned Video World Models for Manipulation. The same action-guided generation paradigm is being extended to robotic manipulation. ContactFlow [11] augments video-based action conditioning with an explicit representation of contact, a physical constraint that purely appearance-driven video world models tend to violate, and shows the representation transfers across embodiments. GeniWorld [107] and DreamX-Phi [336] both predict future observations directly from an observed frame, a language instruction, and a proposed action sequence, differing mainly in how tightly they couple visual realism to the faithfulness of the predicted contact and gripper state. Together, these systems suggest that action-conditioned video generation is converging on a shared recipe across driving and manipulation, with the physical plausibility of the interaction, not the appearance of the rollout, as the binding constraint.

SA-WAM [104] introduces metric depth into joint world–action prediction, denoising future RGB-D observations and robot actions with one backbone. A nonlinear depth encoding reuses the frozen video tokenizer, making geometry available without a separate geometry-specific encoder. Its simulation and real-robot evaluations connect future-state prediction to manipulation success, although better image or depth reconstruction alone remains insufficient evidence of a better policy.

Forecasting-Driven Action Planning. Another line of work forecasts future states from current observations and ego actions, letting planners evaluate outcomes before committing [191, 352, 474]. Different from purely reactive schemes, these approaches emphasize *anticipatory decision-making*, allowing the agent to virtually "test" multiple futures and avoid unsafe trial-and-error in the real world. Drive-WM [368] generates video rollouts of candidate maneuvers, scoring them with image-based rewards for trajectory selection. DriveDreamer [365] proposed the ActionFormer to predict future states and ego-environment interactions. ADriver-I [152] combines multimodal LLMs with autoregressive control signals and world evolution prediction. Vista [98] incorporates uncertainty-aware reward modules for robust action evaluation.

GPT-style designs such as DrivingGPT [54] and DrivingWorld [134] model visual and action tokens jointly for planning via next-token prediction. Integrated frameworks like Doe-1 [471] unify perception, prediction, and planning for closed-loop autonomous driving, while VaVAM [19] bridges video diffusion and an action expert for decision-making. GenAD [470] instead casts generative end-to-end driving as future-scene prediction, learning to forecast and plan jointly from large-scale video. ProphetDWM [362] further couples latent action learning with state forecasting for long-term planning, and DeepSight [450] predicts compact latent states to enable long-horizon world modeling for end-to-end driving. Pushing this paradigm to its extreme, DriveVA [221] repurposes general video-action models as zero-shot drivers without task-specific finetuning. Overall, by simulating diverse futures and leveraging feedback, forecast-driven models enhance generalization and safety in end-to-end autonomous driving.

Forecasting in Geometry Rather Than Pixels. A pointed critique of the pixel-space convention has emerged from within the action-conditioned literature itself. GeoWAM [242] observes that pixels represent scene dynamics only indirectly, entangling geometry and motion with appearance, texture, and illumination, and therefore forcing the model to infer three-dimensional transformations from two-dimensional evidence; it moves the prediction target into a visual-geometry space instead. 4DGS-WAM [249] reaches a compatible conclusion from the representation side: it adopts an explicit 4D Gaussian Splatting scene in which dynamic objects and static background are modeled separately, then predicts per-object Gaussian transformations rather than re-rendering the whole frame, which avoids repeatedly reprocessing redundant background content. WALL-SS [452] addresses the orthogonal axis of horizon, scaling long-horizon prediction through next-scale autoregression. These works matter for this survey's taxonomy because they relocate the forecasting target from the image plane to a native 3D or 4D representation while keeping the action-conditioned interface intact — precisely the boundary between VideoGen and the occupancy- and LiDAR-based branches of Sec. 3.2 and Sec. 3.3.

PointZero [81] instead learns transferable dynamics by completing dense future 3D point tracks from an RGB-D observation and sparse partial tracks. Post-training replaces partial-track conditioning with end-effector poses for robot-conditioned prediction, or adds an action head for imitation learning. These tracked scene points describe motion rather than LiDAR returns, so we discuss the method at this geometric forecasting boundary instead of classifying it as a LiDAR sensor generator.

Direct forecasting can also avoid rendering future RGB altogether. GeoWM [276] predicts depth at a requested horizon from geometric history and a forecast camera viewpoint, whereas VGGTWorld-VLA [232] conditions future geometry tokens on driving semantics and a supplied ego trajectory. Their decoded point maps remain camera-derived geometry rather than LiDAR measurements. These methods extend the geometric forecasting interface without establishing closed-loop control performance; their reference and alignment protocols are discussed in Sec. 4.2.2.

Table 2 Summary of video-based generation (**VideoGen**) models.

- **Datasets:** N nuScenes [32], K KITTI [102], W Waymo Open [328], Y OpenDV-YouTube [418], A Argoverse 2 [378], N nuPlan [33], N NAVSIM [65], C CARLA [78], and P Private (Internal) Data.
- **Input & Output:** Noise Latent, Video (Single-View and/or Multi-View), and Ego-Action.
- **Architectures (Arch.):** *AR*: Autoregressive Models, *MLLM*: Multimodal Large Language Models, *SD*: Stable Diffusion Models, *DiT*: Diffusion Transformer, *GPT*: Generative Pre-trained Transformer.
- **Tasks:** *VG*: Video Generation, *E2E*: End-to-End Planning, and *3SR*: 3D Scene Reconstruction.
- **Categories:** ❶ Data Engine (Sec. 3.1.1), ❷ Action Interpreter (Sec. 3.1.2), and ❸ Neural Simulator (Sec. 3.1.3).

#	Model	Venue	Dataset	Input	Output	Condition	Len.	Freq.	Arch.	Task	Cat.	URL
1	GAIA-1 [127]	arXiv'23	P	Video	Video	T C T	L	25Hz	AR	VG	❷	
2	ADriver-I [152]	arXiv'23	N P	Video	Video Ego	T	S	2Hz	MLLM	VG	❷	
3	BEVControl [420]	arXiv'23	N	Noise	Video	3 H	S	-	SD	VG	❶	
4	MagicDrive3D [95]	arXiv'24	N	Noise	Video	3 B C T	S	12Hz	SD	VG, 3SR	❶ ❸	
5	Delphi [245]	arXiv'24	N	Video	Video	3 H T	L	2Hz	SD	VG	❶	
6	BEVWorld [457]	arXiv'24	N C	Noise	Video	T	S	12Hz	AR	VG	❶	
7	DiVE [154]	arXiv'24	N	Video	Video	3 H T	L	12Hz	DiT	VG	❶	
8	DreamForge [253]	arXiv'24	N	Video	Video	3 H C T	L	12Hz	SD, DiT	VG	❸	
9	SyntheOcc [190]	arXiv'24	N	Noise	Video	P	S	2Hz	SD	VG	❶	
10	HoloDrive [386]	arXiv'24	N	Video	Video	3	S	-	SD	VG	❶	
11	InfinityDrive [114]	arXiv'24	Y N	Video	Video	T T	L	10Hz	AR	VG	❷	
12	UniMLVG [48]	arXiv'24	Y N W A	Video	Video	3 B C T	L	12Hz	DiT	VG	❶	
13	DrivePhysica [432]	arXiv'24	N	Noise	Video	3 H C	L	12Hz	DiT	VG	❶	
14	Doe-1 [471]	arXiv'24	N	Noise	Video Ego	P	S	2Hz	MLLM	VG, E2E	❷ ❸	
15	OccScene [184]	arXiv'24	N K	Video	Video	T T	-	2Hz	SD	VG	❶	
16	DrivingGPT [54]	arXiv'24	N N	Video	Video Ego	T	L	10Hz	GPT	VG, E2E	❷	
17	DrivingWorld [134]	arXiv'24	P N	Video	Video Ego	T	L	10Hz	GPT	VG, E2E	❷	
18	Panacea [375]	CVPR'24	N	Video	Video	3 H C T	S	2Hz	SD	VG	❶	
19	Drive-WM [368]	CVPR'24	N	Video	Video Ego	3 H T T	S	2Hz	SD	VG, E2E	❷	
20	GenAD [418]	CVPR'24	Y	Video	Video Ego	T T	S	2Hz	SD	VG, E2E	❷	
21	DriveDreamer [365]	ECCV'24	N	Video	Video Ego	3 H T T	S	2Hz	SD	VG, E2E	❷	
22	DrivingDiffusion [195]	ECCV'24	N	Video	Video	3 T	S	2Hz	SD	VG	❶	
23	WoVoGen [239]	ECCV'24	N	Noise	Video	B T C T P	S	2Hz	SD	VG	❶	
24	MagicDrive [97]	ICLR'24	N	Noise	Video	3 B C T	S	12Hz	SD	VG	❶	
25	Vista [98]	NeurIPS'24	Y N	Video	Video Ego	T	L	10Hz	SD	VG, E2E	❷	
26	SimGen [488]	NeurIPS'24	N	Noise	Video	3 B	S	2Hz	SD	VG	❶	
27	BEVGen [330]	RA-L'24	N A	Video	Video	B	S	-	AR	VG	❶	
28	VaViM [19]	arXiv'25	Y	Video	Video	N/A	L	2Hz	MLLM	VG	❷	
29	VaVAM [19]	arXiv'25	Y N N	Video	Video Ego	T	L	2Hz	MLLM	VG, E2E	❷	
30	MiLA [351]	arXiv'25	N	Video	Video	T C T	L	12Hz	DiT	VG	❷	
31	GAIA-2 [307]	arXiv'25	P	Video	Video	T C T	L	30Hz	DiT	VG	❷	
32	Nexus [489]	arXiv'25	N W P	Video	Video	B	S	2Hz	DiT	VG	❶ ❸	
33	DriVerse [194]	arXiv'25	N W	Video	Video	T	L	12Hz	DiT	VG	❷	
34	GeoDrive [43]	arXiv'25	N	Video	Video	C	L	12Hz	DiT	VG, 3SR	❶ ❸	
35	Challenger [408]	arXiv'25	N	Noise	Video	3 B T	L	12Hz	DiT	VG	❶	
36	LongDWM [363]	arXiv'25	N	Video	Video	T	L	10Hz	DiT	VG	❷	
37	Cosmos-Drive [302]	arXiv'25	P	Video	Video	3 H T	L	-	DiT	VG	❶	
38	OmniNWM [183]	arXiv'25	N	Video	Video Ego	T C P	L	12Hz	DiT	VG, E2E	❸	
39	WorldSplat [495]	arXiv'25	N	Noise	Video	3 B T T	S	12Hz	DiT	VG, 3SR	❶ ❸	
40	DriveDreamer-2 [464]	AAAI'25	N	Noise	Video	3 H T	S	12Hz	SD	VG	❶	
41	SubjectDrive [136]	AAAI'25	N	Video	Video	3 H T	S	2Hz	SD	VG	❶	
42	DriveScape [381]	CVPR'25	N	Noise	Video	3 H T C T	S	10Hz	SD	VG	❶	
43	DriveDreamer4D [463]	CVPR'25	W	Video	Video	3 H	S	-	SD	VG, 3SR	❶ ❷	
44	DrivingSphere [409]	CVPR'25	N	Noise	Video	P	L	12Hz	DiT	VG	❸	
45	UniScene [181]	CVPR'25	N	Noise	Video	T P	L	12Hz	SD	VG	❶	
46	GEM [118]	CVPR'25	Y N	Video	Video	T	L	10Hz	SD	VG	❷	
47	MaskGWM [274]	CVPR'25	Y N W	Video	Video	T T	L	10Hz	DiT	VG	❷	
48	UMGen [385]	CVPR'25	W N	Video	Video	B T	L	2Hz	AR	VG, E2E	❸	
49	PerLDiff [447]	ICCV'25	N K	Noise	Video	3 H	S	-	SD	VG	❶	
50	DriveArena [423]	ICCV'25	N	Video	Video	3 H C T	L	12Hz	SD, DiT	VG	❶ ❸	
51	MagicDrive-V2 [96]	ICCV'25	N W	Noise	Video	3 B T C T	L	12Hz	DiT	VG	❶	
52	InfiniCube [241]	ICCV'25	W	Video	Video	H P	L	10Hz	SD	VG, 3SR	❸	
53	DiST-4D [113]	ICCV'25	N	Video	Video	3 B T C	L	12Hz	DiT	VG, 3SR	❸	
54	Epona [449]	ICCV'25	N N	Video	Video Ego	T	L	5Hz	DiT	VG, E2E	❷	
55	AutoScape [46]	ICCV'25	N A	Video	Video	3 H T T	L	12Hz	SD	VG, 3SR	❶	
56	Glad [393]	ICLR'25	N	Video	Video	3 H T	S	2Hz	SD	VG	❶	
57	DualDiff [188]	ICRA'25	N	Noise	Video	3 H C T P	S	-	SD	VG	❶	
58	PosePilot [160]	IROS'25	N	Video	Video	C	L	2Hz	SD,DiT,AR	VG	❷	
59	STAGE [353]	IROS'25	N	Video	Video	3 H T	L	12Hz	SD	VG	❶	
60	OmniDrive [256]	arXiv'26	N	Noise	Video	3 H T C T	S	12Hz	DiT	VG	❶	
61	OmniDreams [279]	arXiv'26	P	Video	Video Ego	T	L	-	DiT	VG, E2E	❸	
62	Xiaomi-WM [478]	arXiv'26	N W P	Video	Video	H T C T	L	30Hz	DiT	VG, 3SR	❸	
63	GenieDrive [431]	CVPR'26	N	Video	Video	C T P	L	12Hz	DiT	VG	❷	
64	GaussianDWM [69]	CVPR'26	N	Video	Video	T C T	S	12Hz	SD	VG, 3SR	❶	
65	UniFuture [206]	ICRA'26	N	Video	Video	N/A	S	12Hz	SD	VG, 3SR	❸	

Table 2 Continued summary of VideoGen models. Symbols and categories follow the preceding part. Text names identify additional datasets or modalities; a dash denotes a temporal field not standardized in this summary. FF denotes feed-forward reconstruction; GS denotes Gaussian splatting. SA-WAM additionally conditions on robot proprioception and generates actions. RoGSW4RLD uses robot kinematics and reconstructs supplied rollouts. DepthWorld uses end-effector poses and a five-frame future window (six history frames); SD includes SVD. World Observer generates joint actor–panorama streams in 77-frame chunks.

#	Model	Venue	Dataset	Input	Output	Condition	Len.	Freq.	Arch.	Task	Cat.	URL
66	SpatialCrafter [86]	arXiv'26	SpatialGen RE10K/DL3DV		RGB-D	C	-	-	DiT	VG, 3SR	❶	
67	4DStreamCtrl [192]	arXiv'26	OpenVidHD-Motion3D			C F D	-	-	DiT	VG	❶	
68	4Director [38]	arXiv'26	RealCOD-Rigid			C S T	-	-	DiT	VG	❶	
69	CoDrive [255]	arXiv'26	CoDrive-Bench real/synthetic			C T	-	-	DiT	VG	❷	
70	HelloWorld (Driving) [236]	arXiv'26	P N			T H 3 T	-	-	DiT	VG	❸	
71	SA-WAM [104]	arXiv'26	RoboCasa LIBERO/UR5	RGB-D	RGB-D Action	T	-	-	DiT	VG	❷	
72	RoGSW4RLD [170]	arXiv'26	DROID		4D GS	C S	-	-	FF	3SR	❸	
73	DepthWorld [18]	CoRL'26	DROID-3D	RGB-D	RGB-D	End-effector pose	5	5Hz	SD	VG	❷	
74	World Observer [61]	arXiv'26	Real panorama CARLA	Perspective Panorama	Perspective Panorama	C D T	77	-	DiT	VG	❸	

3.1.3 Neural Simulators

Closed-loop simulators produce realistic virtual worlds that support effective planning, decision-making, and interaction. Regarding the difference in scene modeling, recent methods can be broadly categorized into two main approaches.

Generation-Driven Simulation. Recent advances in generative simulators for autonomous driving leverage conditional generative frameworks [27, 286, 305] to create interactive high-fidelity environments. DriveArena [423] establishes the first closed-loop framework through two core components: TrafficManager for scalable traffic synthesis and WorldDreamer for autoregressive scene generation. Building on this foundation, DreamForge [253] enhances long-term scenario modeling by integrating object-wise position encoding, supported by a novel temporal attention mechanism. Further extending these capabilities, DrivingSphere [409] introduces 4D semantic occupancy modeling that unifies static environments and dynamic objects, coupled with a visual synthesis module ensuring spatiotemporal consistency in multiview video generation. UMGen [385] simulates behavioral interactions between ego-vehicles and user-defined agents, while Nexus [489] dynamically updates environments based on agent decisions, rigorously validated through nuPlan closed-loop benchmarks. GeoDrive [43] advances trajectory optimization for VLA systems via geometry-aware scene modeling and precision control modules. OmniDreams [279] targets real-time generative simulation for closed-loop autonomous driving, while the Xiaomi Auto World Model [478] jointly reconstructs and generates driving scenes within a unified framework. ReSim [419] emphasizes *reliable* world simulation, improving action controllability and the faithfulness of rollouts to commanded maneuvers, and HERMES++ [486] extends the HERMES line into a unified driving world model that couples 3D scene understanding with generation. Collectively, these developments transition generative simulation from passive environment rendering to closed-loop systems capable of agent interaction and feedback-driven adaptation. Recent systems further emphasize real-time, interactive rollouts: CausalDrive [412] builds a real-time causal driving world model for non-oracle closed-loop simulation, and Point-as-Skeleton [380] conditions autoregressive generation on accumulated point clouds to stabilize long closed-loop rollouts. From Pixels to States [200] steps back from this cluster to ask what an interactive world model needs to expose to feel like a game engine, arguing pixel-accurate rollout is not the same target as a coherent, queryable game state. Streaming Multi-Agent Autoregressive Diffusion [265] and Population-Scalable Multi-Agent World Modeling [466] both push closed-loop simulation toward a variable and potentially large number of interacting agents, maintaining one persistent world state rather than one state per agent. Wonder [400] targets real-time, camera-controllable exploration of a generated world, and a video-representation regularization technique [51] directly attacks the compounding error that degrades all of these rollouts over long horizons.

World Observer [61] addresses out-of-view dynamics by jointly generating a perspective actor and panoramic observer streams grounded in shared geometry. Observer Sink references preserve appearance when objects

re-enter, while separate observer prompts control unseen evolution. Its real and synthetic benchmarks assess exit-and-re-entry behavior and estimated 3D displacement within a single generation chunk, under panoramic conditioning. WorldCrafter [439] uses a pose-guided readout of selected historical views to form a fixed set of target-view memory tokens without explicit depth correspondences, supporting camera-controlled streaming exploration and scene revisits. Recent temporal context supports motion continuity, while few-step distillation accelerates generation.

The driving system HelloWorld [236] combines a block-causal multi-view generator, adaptation to self-generated histories, and few-step distillation for repeated observation synthesis. Ego poses, HD maps, and 3D boxes control synchronized camera outputs, followed by conditional LiDAR synthesis. Its causal interface concerns access to observation history; trajectories remain externally supplied, and the system does not itself propagate vehicle dynamics from low-level controls. This distinction separates a controllable sensor simulator from a complete interactive dynamics model. This driving system is distinct from the social-interaction HelloWorld [283] discussed below.

Reconstruction-Centric Simulation. Reconstruction-based simulators employ neural scene reconstruction techniques such as NeRF [259] and 3D GS [165] to convert driving logs into interactive neural environments [55, 85, 90, 100, 139, 155, 184, 187, 251, 266, 287, 303, 440, 455, 477, 497]. StreetGaussian [414] represented dynamic urban street as a set of point clouds equipped with semantic logits and 3D Gaussians, each associated with either a foreground vehicle or the background. Other key implementations include HUGSIM [476], which integrates physical constraints with 3D GS for aggressive behavior synthesis, alongside frameworks like UniSim [181] and Uni-Gaussians [441] that generate synchronized multi-modal sensor outputs through Gaussian primitive distillation. OmniRe [58] further enhances dynamic entity modeling via neural scene graph representations. While conventional 3D GS methods [95, 206, 238, 241, 414] struggle with viewpoint extrapolation artifacts, emerging solutions integrate 3D scene generation models as the data foundation to improve reconstruction robustness. ReconDreamer [273] applies progressive refinement to eliminate ghosting effects in dynamic scenes, while Stage-1 [356] achieves controllable 4D synthesis through multiview point cloud completion. These generation-enhanced approaches [113, 273, 415, 463, 465] demonstrate significant improvements in handling novel viewpoints, effectively bridging the fidelity gap between simulated and real-world environments. More recently, WorldSplat [495] adopts a Gaussian-centric feed-forward design for 4D scene generation, and GaussianDWM [69] builds a 3D Gaussian driving world model that unifies scene understanding and multi-modal generation. Complementary efforts improve robustness to imperfect inputs and viewpoint change: Cam2Sim [150] reconstructs closed-loop simulators from camera logs, CARLA-GS [138] decouples representation, reasoning, and physics simulation for corner-case synthesis, and Adaptive Gaussian Graph [74] performs 4D driving reconstruction in the wild from noisy or generated priors. In parallel, feed-forward geometry models are emerging as a scalable front-end for such reconstruction, regressing 3D structure directly from uncalibrated images – *e.g.*, Glob3R [68] for global feed-forward structure-from-motion, NoDrift3R [329] for drift-robust unposed reconstruction, and OmniX [157] for any-view, any-time 4D reconstruction.

RoGSW4RLD [170] extends reconstruction-centric modeling to predicted robot futures. It jointly lifts synchronized multi-camera rollouts into a metric 4D Gaussian field using articulated robot geometry and kinematics, then refines geometry and appearance while preserving the inferred temporal displacements. The output is queryable across viewpoints and time, but reconstructs a generator's visual future rather than learning an independent geometric transition model; consistency of the lifted scene therefore remains distinct from correctness of the predicted interaction.

Beyond Driving: Social and Surgical Simulation. Generation-driven simulation is also diversifying by application. HelloWorld [283] extends the interactive world model to support social interaction with characters inside the generated world, a capability orthogonal to physical fidelity. In the surgical domain, CrossScope [225] models a dual-scope endoscopic setting with two independently moving observers, and Surg-UniWorld [480] unifies multimodal control for synthesizing instrument–tissue interactions, both adapting the closed-loop simulator paradigm to a setting where the "ego" viewpoint is not a vehicle.

Simulators as Physical and Behavioural Oracles. A further shift concerns what a neural simulator is expected to be faithful *to*. SPVC [186] targets the failure that limits reconstruction-centric simulation in practice: rendered views degrade under extrapolated ego trajectories and scene edits, producing blurry structure,

temporal flicker, and foreground–background misalignment, and it addresses this with a structured, panoptic video-fixing stage that transfers across datasets rather than being tuned per setting. 4DSynth [296] couples procedural generation with 4D generative dynamics, turning a language description, blueprint mask, or single photograph into an editable environment that carries explicit geometry, animated actors, collision-free trajectories, and physics-ready state — that is, a simulator whose output is a simulable scene rather than a video. BehaviorWorldGen [354] closes the remaining loop between action models and controllable world simulators, and CLAP [219] reports that cross-embodiment video world models can act as zero-shot physical simulators. Taken together, these results move the acceptance criterion for a neural simulator from perceptual realism toward physical simulability and behavioural response, which is the same criterion the evaluation protocols of Sec. 4.2.6 are being built to test.

Summary & Insights. Across VideoGen, the field has moved from BEV-conditioned single-frame synthesis to long-horizon, multi-view, and increasingly closed-loop generation, with diffusion transformers and geometry-aware conditioning now dominant. Three limitations recur: (i) maintaining *3D and temporal consistency* over long rollouts, where error accumulation still induces flicker, geometric drift, and appearance/identity changes; (ii) *object-level controllability* – fine-grained control of pose, motion, and interaction remains harder than global appearance or style control; and (iii) a trade-off between the two simulator families – reconstruction-centric simulators inherit metric geometry from real logs but struggle with wide viewpoint extrapolation, whereas generation-driven simulators are flexible yet must actively suppress drift. Promising directions include hybrid generation–reconstruction pipelines, explicit memory and history re-anchoring, and physics- or geometry-grounded conditioning to close the fidelity gap for closed-loop use.

3.2 World Modeling from Occupancy Generation

Generation models based on occupancy grids are tailored to offer a geometry-centric representation that encodes both semantic and structural details of the 3D world. By generating, forecasting, or simulating occupancy in 3D/4D space, these models provide a geometry-consistent scaffold for perception, enable action-contingent future prediction, and support realistic large-scale simulation. Based on their primary function, existing methods can be grouped into **three** categories: ❶ **Scene Representors**, ❷ **Occupancy Forecasters**, and ❸ **Autoregressive Simulators**. Table 3 summarizes existing models under these domains.

3.2.1 Scene Representors

Occupancy-based 3D and 4D generation models, designed for learning structured 3D scene representations, treat the occupancy grid as a geometry-consistent intermediate for downstream tasks. Such a paradigm enhances perception robustness and provides structural guidance for 3D scene generation across two main applications.

3D Perception Robustness Enhancement. Occupancy-based representations have emerged as a powerful intermediate modality for enhancing perception robustness through generative modeling techniques. SSD [179] pioneered this direction by employing discrete [10] and latent diffusion [305] models for scene-level 3D categorical data generation, learning to map sparse occupancy inputs into dense semantic reconstructions. SemCity [180] further improves geometric and semantic fidelity by conditioning diffusion on initial SSC outputs, reducing inconsistencies in reconstructed scenes. Occ-

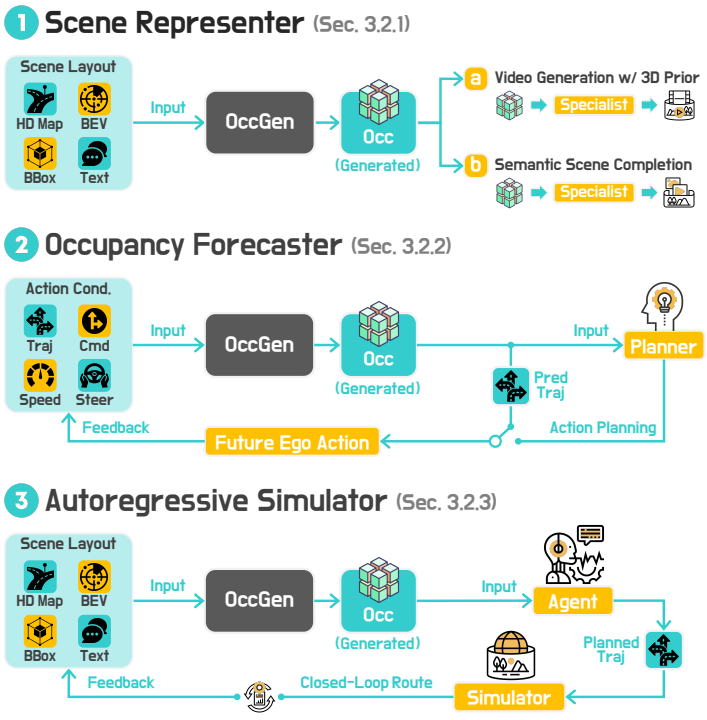


Figure 5 The categorization of **OccGen** models based on functionalities, including scene representors (Sec. 3.2.1), forecasters (Sec. 3.2.2), and autoregressive simulators (Sec. 3.2.3).

Table 3 Summary of occupancy-based generation (**OccGen**) models.

- **Datasets:** S SemanticKITTI [21], C CarlaSC [379], N Occ3D-nuScenes [339], W Waymo Open [328], L Lyft-Level5 [125], A Argoverse 2 [378], 3 KITTI-360 [208], U NYUv2 [321], and O OpenCOOD [403].
- **Input & Output:** Noise, Latent Codebook, Images, 3D Occ, 4D Occ, and Ego-Action.
- **Architectures (Arch.):** *Enc-Dec*: Encoder-Decoder, *LDM*: Latent Diffusion Model, *DDM*: Discrete Diffusion Model, *MSSM*: Memory State-Space Model, *AR*: Autoregressive Model, *DiT*: Diffusion Transformer, *LLM*: Large Language Model.
- **Tasks:** *O3G*: 3D Occupancy Generation, *O4G*: 4D Occupancy Generation, *OF*: 4D Occupancy Forecasting, *PT*: Pre-Training, *SSC*: Semantic Scene Completion, and *E2E*: End-to-End Planning.
- **Categories:** ❶ Scene Representor (Sec. 3.2.1), ❷ Occ Forecaster (Sec. 3.2.2), and ❸ AR Simulator (Sec. 3.2.3).

#	Model	Venue	Dataset	Input	Output	Condition	Len.	Arch.	Task	Cat.	URL
1	Emergent-Occ [166]	ECCV'22	N			N/A	7	Enc-Dec	OF, E2E	❷	
2	SSD [179]	arXiv'23	C			N/A	1	LDM	O3G	❶	
3	UniWorld [263]	arXiv'23	N			N/A	-	Enc-Dec	PT	❷	
4	FF4D [167]	CVPR'23	S N A			N/A	5	Enc-Dec	OF	❷	
5	UrbanDiff [448]	arXiv'24	N			B	1	LDM	O3G	❶	
6	OccSora [355]	arXiv'24	C N W			T	32	DiT	O4G	❷ ❸	
7	LOPR [178]	arXiv'24	N W			T	15	Enc-Dec	OF	❷	
8	OccLLaMA [373]	arXiv'24	N			T	6	LLM	O4G	❷	
9	FSF-Net [112]	arXiv'24	N			N/A	4	Enc-Dec	OF, E2E	❷	
10	DOME [109]	arXiv'24	N			T	11	DiT	OF	❷	
11	GaussianAD [472]	arXiv'24	N			N/A	6	Enc-Dec	OF, E2E	❷	
12	OccScene [184]	arXiv'24	N S U			N/A	1	SD	O3G	❶	
13	DFIT-OccWorld [444]	arXiv'24	N			T	6	Enc-Dec	OF, E2E	❷	
14	Cam4DOcc [246]	CVPR'24	N L			T	4	Enc-Dec	OF	❷	
15	XCube [301]	CVPR'24	W			N/A	1	LDM	O3G	❸	
16	SemCity [180]	CVPR'24	S C			N/A	1	LDM	O3G, SSC	❶ ❸	
17	DriveWorld [262]	CVPR'24	N			V S	4	MSSM	OF, PT	❷	
18	UnO [3]	CVPR'24	S N A			N/A	6	Enc-Dec	OF	❷	
19	PDD [230]	ECCV'24	S C			N/A	1	DDM	O3G	❸	
20	OccWorld [469]	ECCV'24	N			T	6	AR	OF, E2E	❷	
21	WoVoGen [239]	ECCV'24	N			B	3	LDM	O4G	❶ ❷	
22	UniScene [261]	RA-L'24	N			N/A	-	Enc-Dec	PT	❷	
23	T^{3}Former [398]	arXiv'25	N			T	6	AR	OF, E2E	❷	
24	COME [318]	arXiv'25	N			T	6	DiT	OF	❷	
25	PrITTI [343]	arXiv'25	3			3 B	1	DiT	O3G	❸	
26	Drive-OccWorld [426]	AAAI'25	N L			C T V S	4	AR	OF, E2E	❷	
27	DrivingSphere [409]	CVPR'25	N			B	-	LDM	O4G	❶ ❸	
28	EfficientOCF [401]	CVPR'25	N L			T	4	Enc-Dec	OF	❷	
29	UniScene [181]	CVPR'25	N			B	6	DiT	O4G, OF	❶ ❸	
30	DIO [71]	CVPR'25	A			N/A	5	Enc-Dec	OF	❷	
31	InfiniCube [241]	ICCV'25	W			3 H	1	LDM	O3G	❶ ❷	
32	Control-3D-Scene [231]	ICCV'25	C			G	1	DDM	O3G	❶	
33	UniOcc [367]	ICCV'25	N C W O			N/A	6	N/A	OF	❷	
34	I^{2}World [209]	ICCV'25	N W			C T V S	6	AR	OF	❷	
35	DynamicCity [24]	ICLR'25	C N W			3 C T	16	DiT	O4G	❸	
36	PreWorld [193]	ICLR'25	N			T	6	Enc-Dec	OF, E2E	❷	
37	OccProphet [47]	ICLR'25	N L			T	4	Enc-Dec	OF	❷	
38	RenderWorld [416]	ICRA'25	N			T	6	AR	OF, E2E	❷	
39	Occ-LLM [404]	ICRA'25	N			T	6	LLM	OF, E2E	❷	
40	$\mathcal{X}$-Scene [425]	NeurIPS'25	N			3 B H	1	LDM	O3G	❶ ❸	
41	OccSim [224]	arXiv'26	N			T	3K+	DiT	O4G	❸	
42	OccDirector [207]	arXiv'26	N			T	-	DiT	O4G	❸	
43	SparseWorld [64]	AAAI'26	N			T	6	Enc-Dec	OF, E2E	❷	
44	EditSSC [14]	CVPRW'26	S			N/A	1	LDM	O3G, SSC	❶	
45	IR-WM [254]	ICRA'26	N			T	6	Enc-Dec	OF, E2E	❷	
46	OccTENS [159]	RA-L'26	N			T	6	AR	O4G, OF	❸	

Gen [350] casts multi-modal 3D occupancy prediction as a generative, coarse-to-fine denoising process that progressively refines the

Table 3 Continued summary of OccGen models. Symbols and categories follow the preceding part. Text names identify additional datasets or modalities; a dash denotes a temporal field not standardized in this summary. LighTROcc uses camera calibration and observed ego poses without commanded future ego actions; its four future steps reach two seconds and cover movable objects.

#	Model	Venue	Dataset	Input	Output	Condition	Len.	Arch.	Task	Cat.	URL
47	InterOCF [453]	arXiv'26	nuScenes-Occ Lyft-Level5	Camera/LiDAR		N/A	-	Enc-Dec	OF	❷	
48	TriO [331]	arXiv'26	A	Radar		N/A	-	Enc-Dec	OF	❷	
49	LighTROcc [162]	arXiv'26	N			N/A	4	Enc-Dec	OF	❷	

occupancy field from camera and LiDAR inputs.

Generation Consistency Guidance. Other works leverage occupancy to guide high-fidelity, temporally coherent scene synthesis. WoVoGen [239] proposes 4D temporal occupancy volumes to drive multi-view video generation with intra-world and inter-sensor consistency. UrbanDiff [448] uses semantic occupancy grids as geometric priors for 3D-aware image synthesis, while DrivingSphere [409] transforms dynamic 4D occupancy scenes into temporally consistent video via semantic rendering. UniScene [181] generalizes occupancy-based generation across modalities, combining Gaussian-based rendering [165] with prior-guided sparse modeling for unified video and LiDAR synthesis. EditSSC [14] further enables editable semantic occupancy scene generation through diffusion. GaussianDWM++ [70] pushes the representation further by grounding a 3D Gaussian driving world model in language, coupling scene understanding with controllable 4D editing rather than treating occupancy as a purely geometric intermediate. Collectively, these methods highlight the role of occupancy grids as a unifying structural prior for producing spatially and temporally consistent outputs with high structural fidelity.

Completion as Generative Inference. Occupancy-style scene completion has traditionally been posed as discriminative voxel labelling, which is poorly matched to its own statistics: an outdoor LiDAR scan observes on the order of one percent of the target volume, under class imbalance exceeding three orders of magnitude. GSSC [49] recasts semantic scene completion as generative inference under a single discrete-diffusion formulation, and uses it in three distinct roles — synthesizing matched sparse-observation/dense-completion pairs to attack the long tail at its source, generating semantically guided completions, and releasing the resulting PS^3-SemanticKITTI corpus for training. This is a notable case in which the generative and reconstructive functional types collapse into one mechanism: the same model supplies the training data and performs the completion.

3.2.2 Occupancy Forecasters

Models for 4D occupancy forecasting predict future occupancy from ego actions and past observations, allowing anticipation of environmental changes. This capability serves two purposes: as a self-supervised pretraining task for building generalizable 3D/4D models, and as a dynamic predictor for behavior-aware, controllable future scene generation.

Predictive Model Pretraining. Several methods explore occupancy forecasting as a pretext task to learn rich spatiotemporal features from LiDAR sequences, building generalizable generation models via self-supervised learning. Emergent-Occ [166, 167] introduces differentiable rendering to reconstruct point clouds from 4D occupancy predictions, enabling self-supervised training from raw sequences. UnO [3] models a continuous 4D occupancy field for joint perception and forecasting. Large-scale pretraining frameworks such as UniWorld [263], UniScene [261], and DriveWorld [262] combine image and LiDAR data to learn foundational occupancy models that can be fine-tuned for downstream tasks like detection and planning, reducing reliance on dense labels while improving generalization.

TriO [331] broadens self-supervised occupancy modeling to camera, LiDAR, and radar observations. It queries occupancy, obstacle probability, and flow in continuous space-time, using ray-based occupancy supervision, image–LiDAR obstacle pseudo-labels, and radar Doppler measurements for flow. A LiDAR forecasting

head evaluates the learned representation through future sensor returns. Comparisons must account for the additional sensor modalities and distinguish depth accuracy from point-cloud coverage rather than treating all forecasting scores as interchangeable.

Ego-Conditioned Occupancy Forecasting. Other approaches forecast occupancy conditioned on both history and ego-agent actions, supporting behavior-aware and controllable prediction. OccWorld [469] jointly models ego motion and surrounding environment evolution in 3D occupancy space, while OccSora [355] generates trajectory-conditioned 4D occupancy over long horizons. GaussianWorld [498] recasts streaming occupancy prediction as Gaussian-based scene evolution, propagating a 3D Gaussian state across frames for efficient online forecasting. Later works enhance controllability [109, 318], fidelity [444], temporal coherence [71, 112, 472], and efficiency [398]. OccTENS [159] casts 3D occupancy world modeling as temporal next-scale prediction, SparseWorld [64] adopts sparse, dynamic queries for 4D occupancy modeling, and IR-WM [254] performs vision-centric 4D occupancy forecasting and planning via implicit residual world models. Recent forecasters further tighten the link to downstream planning: ForecastOcc [267] predicts future semantic occupancy directly from vision, while the Gaussian Evolution Model (GEM) [44] couples occupancy forecasting with motion planning under an evolving Gaussian representation. Vision-centric pipelines like Cam4DOcc [246] and its successors [416, 426] integrate world models into end-to-end planning to empower their generative abilities. OccLLaMA [373] and Occ-LLM [404] unify vision, language, and action modalities with semantic occupancy as the shared representation to support embodied question answering, while UniOcc [367] establishes a benchmark combining real and simulated data for standardized evaluation. Together, these works position occupancy forecasting as both a powerful self-supervised learning objective and a key tool for modeling dynamic, action-contingent world states. Recent occupancy world models further tighten the coupling to planning and representation: OWMDrive [59] drives causality-aware end-to-end planning from a 4D occupancy world model, while CascadeOcc [147] rethinks occupancy world-model tokenization with cascaded vector-quantized representations for forecasting and planning without external modalities. Two further works target the 2D-to-3D lifting step that most vision-centric forecasters depend on: InterOCF [453] models spatio-temporal 2D-3D interaction explicitly for camera-only 4D occupancy forecasting, and a geometry-aware context-modeling approach [50] targets the geometric distortion that autoregressive occupancy tokenization tends to introduce in static structures.

LighTROcc [162] predicts movable-object occupancy from surround-view images using compact instance queries and anisotropic 3D Gaussian mixtures. Learned displacements propagate each instance across future horizons in one forward pass. Its voxel and instance evaluations cover up to two seconds, with results reported under supplemented occupancy labels and an additional protocol for the original sparse annotations. This representation targets movable objects without commanded ego-action conditioning; its annotation-specific scores should not be merged with full-scene forecasting results.

3.2.3 Autoregressive Simulators

The occupancy-based autoregressive simulators generate large-scale, temporally coherent 4D occupancy for realistic and interactive simulation. They serve as foundation simulators for perception, planning, and decision-making, with research focusing on two directions: generating scalable unbounded environments and modeling long-horizon dynamics for controllable closed-loop simulation.

Scalable Open-World Generation. Coarse-to-fine and outpainting strategies have been explored to construct large-scale, unbounded 3D occupancy environments. PDD [230] proposes a scale-varied diffusion framework that progressively generates outdoor scenes from coarse layouts to fine details, while XCube [301] adopts hierarchical voxel-based latent diffusion for multi-resolution generation. SemCity [180] adds manipulation functions for scene editing, and InfiniCube [241] and $\mathcal{X}$-Scene [425] integrate voxel-based occupancy with consistent visual synthesis for realistic, editable simulation worlds. Together, these works construct scalable occupancy-based representations that serve as interactive and extensible environments for embodied agents.

Long-Horizon Dynamic Simulation. Other works focus on autoregressive 4D occupancy generation to simulate dynamic world evolution. OccSora [355] produces trajectory-conditioned sequences over 16-second horizons, while DynamicCity [24] enables layout-aware and command-conditioned generation, supporting controllable scene synthesis and agent interaction. DrivingSphere [409] constructs a 4D world comprising static backgrounds and dynamic objects for closed-loop simulation, and UniScene [181] generates layout-conditioned 4D occupancy

with rich semantic and geometric detail. OccSim [224] pushes occupancy world-model simulation to multi-kilometer long horizons, while OccDirector [207] supports language-guided behavior and interaction generation in 4D occupancy space. AutoWorld [293] scales multi-agent traffic simulation by learning a self-supervised world model that jointly rolls out the behaviors of many interacting agents. These approaches integrate spatial structure and temporal coherence to create realistic, controllable environments for embodied agent simulation and decision-making.

Summary & Insights. OccGen methods exploit the geometry-consistent, semantically labeled voxel grid as a structural scaffold, spanning scene representors, occupancy forecasters, and autoregressive simulators. Their shared bottleneck is the *cubic memory and compute cost* of dense 3D/4D voxels, which forces aggressive spatial–temporal compression (triplane, sparse, or Gaussian latents) and caps achievable resolution and horizon; *class imbalance and long-tail geometry* (thin structures, small dynamic agents) further erode fine detail, and long-horizon autoregressive forecasting still accumulates error. The most promising directions are efficient sparse/latent representations, explicit handling of rare classes, and tighter coupling of occupancy forecasting with downstream planning.

3.3 World Modeling from LiDAR Generation

LiDAR-based generation models provide geometry-aware and appearance-invariant representations by modeling complex scenes from point clouds. They enable robust 3D scene understanding and high-fidelity geometric simulation, offering advantages over image- and occupancy-based approaches in both geometric fidelity and environmental robustness. Based on their primary function, these methods can be classified into **three** categories: ❶ **Data Engines**, ❷ **Action Interpreters**, and ❸ **Autoregressive Simulators**. Table 4 summarizes existing models under these domains.

3.3.1 Data Engines

LiDAR-based data engines mitigate the scarcity of large-scale LiDAR training data due to high acquisition costs and annotation challenges by generating diverse and controllable point clouds [173, 201]. Such models enhance perception robustness, enable geometrically accurate scene completion, and support the synthesis of rare or cross-modal scenarios [205]. Recent approaches focus on four major applications.

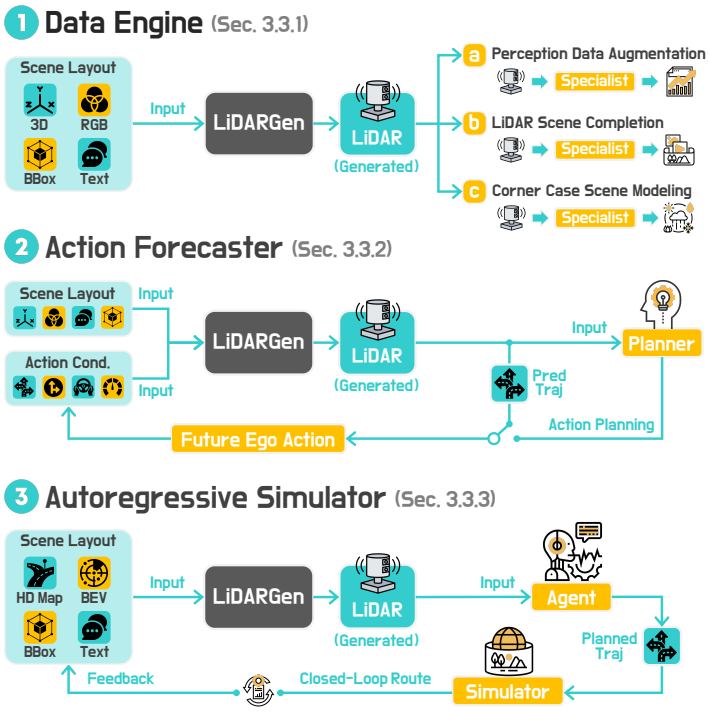


Figure 6 The categorization of **LiDARGen** models based on functionalities, including data engines (Sec. 3.3.1), action forecasters (Sec. 3.3.2), and autoregressive simulators (Sec. 3.3.3).

Perception Data Augmentation. LiDAR-based generative modeling supports data augmentation for core 3D perception tasks such as detection and segmentation, with an emphasis on geometric fidelity and sensor realism. Early approaches primarily focused on modeling uncertainty and spatial structure to synthesize realistic LiDAR scans. DUSty [268] is a GAN-based framework that synthesizes realistic LiDAR scans by explicitly disentangling the underlying depth map from measurement uncertainty. DUSty v2 [270] extends DUSty by incorporating implicit neural representations, enabling the model to generate LiDAR range images at arbitrary resolutions. LiDARGen [499] pioneered the application of Langevin dynamics for LiDAR point cloud generation, achieving superior performance compared to GANs and VAEs. As the first work to adopt the denoising-diffusion paradigm in this domain, it has inspired numerous subsequent studies based on Denoising

Diffusion Probabilistic Models (DDPMs) [123]. With explicit positional encoding, R2DM [269] achieves higher-precision LiDAR point cloud generation through a standardized DDPM process. Leveraging flow matching [214], R2Flow [271] significantly accelerates LiDAR point cloud generation.

LiDM [300], RangeLDM [130], and 3DiSS [278] adopt latent diffusion technology by first compressing raw-scale data into low-dimensional latent variables through a pretrained VAE, then training the diffusion model in this latent space. The generated outputs are reconstructed to the original resolution, substantially improving generation speed while preserving quality. LiDARGRIT [116] extends this paradigm by discretizing the latent space with VQ-VAE [345] and generating latent codes using an autoregressive transformer. It further introduces a raydrop estimation loss to explicitly enhance the raydrop noise modeling. SDS [87] proposes simultaneous diffusion sampling for multi-view LiDAR scene generation, producing all views together to achieve much better geometric consistency than generating each view separately. Recently, SPIRAL [490] pioneered the generation of segmentation-labeled LiDAR data and introduced a novel closed-loop inference strategy that enhances consistency between geometry and semantics. La La LiDAR [229] proposes a layout-guided generative framework that integrates scene graph-based layout diffusion with a foreground-aware control injector, enabling explicit modeling of object relations and controllable scene generation. Veila [227] introduces a conditional diffusion framework for panoramic LiDAR generation guided by a monocular RGB image. It addresses the challenges of reliable conditioning, cross-modal alignment, and maintaining structural coherence beyond the RGB field of view. Several recent diffusion variants further improve fidelity and controllability: TopoLiDM [216] injects topology awareness for interpretable and realistic point-cloud generation, SG-LDM [391] performs semantic-guided generation via latent-aligned diffusion, R3DPA [313] transfers RGB-pretrained priors through 3D representation alignment, and L3DR [223] couples 3D-aware diffusion with an explicit rectification stage. OmniLiDAR [228] unifies multi-domain 3D LiDAR generation within a single diffusion framework, while LiDARDraft [372] enables controllable LiDAR point-cloud generation from versatile inputs. These advances enhance LiDAR-based perception by generating diverse, controllable, and geometrically faithful training data that capture real-world sensing characteristics. Continuing this trend, T2LDM++ [298] advances text-to-LiDAR scene generation with self-conditioned representation guidance, and adversarially guided diffusion [29] improves the realism of synthesized LiDAR range images.

Scene Completion. The completion of 3D scenes aims to reconstruct dense and coherent 3D geometry from sparse or occluded LiDAR scans, with recent generative methods improving geometric fidelity and controllability. UltraLiDAR [397] introduces a discrete voxel-based representation for LiDAR point clouds using a VQ-VAE [345], enabling efficient and controllable sparse-to-dense completion. LiDiff [277] and DiffSSC [37] utilize the denoising process of DDPM to reposition duplicated points, thereby densifying the LiDAR point cloud while simultaneously completing occluded areas. Building on UltraLiDAR [397] for background completion and AnchorFormer [57] for foreground object synthesis, LiDAR-EDIT [124] enables flexible editing of LiDAR scenes, including object removal and insertion. LiDPM[252] revisits LiDiff [277] with global DDPM denoising for scene completion, and demonstrates unconditional generation by disabling scan conditioning and initializing from point clouds sampled from arbitrary shapes. Similarly, Distillation-DPO[460] enhances both completion quality and inference efficiency of LiDiff [277] through the integration of Score Distillation[292] and Diffusion-DPO[348]. Recently, SuperPC [79] proposes a unified framework that transforms point clouds into representation features suitable for completion, upsampling, denoising, and colorization, thereby avoiding the error accumulation that can arise from sequentially applying separate models. More recent completion models scale and accelerate this process: PointDiffusion [2] performs single-step latent diffusion for point-cloud scene completion, while PatchScene [402] adopts patch-based voxel diffusion with spatio-temporal fusion for large-scale completion.

Rare Condition Modeling. To improve the robustness of 3D perception in adverse conditions, recent methods explore controllable LiDAR generation for safety-critical scenarios. Text2LiDAR [383] presents a Transformer-based architecture that integrates textual information to enable text-controlled LiDAR point cloud generation. WeatherGen [384] targets rainy, snowy, and foggy conditions, generating high-quality LiDAR point clouds for these conditions within a unified controllable generative model. The practical utility of the generated point cloud data is validated through 3D object detection tasks in these adverse weather scenarios. OLiDM [411] addresses fidelity limitations at the object level via a two-stage pipeline: it first generates foreground objects, which are then used as conditions for scene generation, ensuring controllable and high-quality results at both object and scene levels. Meanwhile, LOGen [172] proposes an object-level point cloud generation model to

Table 4 Summary of LiDAR-based generation (**LiDARGen**) models.

- **Datasets:** K KITTI [102], S SemanticKITTI [21], N nuScenes [32], 3 KITTI-360 [208], P PandaSet [392] C Carla [78], S SeeingThroughFog [25], W Waymo [328], N NAVSIM [65], A Argoverse 2 [378] and O OmniDrive [358].
- **Input & Output:** Noisy Latent, Latent Codebook, Noisy LiDAR Point Cloud, LiDAR Point Cloud, LiDAR Sequence, and Images/Videos (Single-View and/or Multi-View).
- **Architectures (Arch.):** *GAN*: Generative Adversarial Network, *Enc-Dec*: Encoder-Decoder, *LDM*: Latent Diffusion Model, *AR*: Autoregressive Model, *DiT*: Diffusion Transformer, *LLM*: Large Language Model.
- **Tasks:** *LG*: LiDAR Generation, *L4G*: 4D LiDAR Generation, *SEG*: 3D Semantic Segmentation, *DET*: 3D Object Detection, *SC*: Scene Completion, *OP*: Occupancy Prediction, and *E2E*: End-to-End Planning.
- **Categories:** ❶ Data Engine (Sec. 3.3.1), ❷ Action Forecaster (Sec. 3.3.2), and ❸ AR Simulator (Sec. 3.3.3).

#	Model	Venue	Dataset	Input	Output	Condition	Len.	Arch.	Task	Cat.	URL
1	DUSty [268]	IROS'21	K			N/A	1	GAN	LG, SC	❶	
2	LiDARGen [499]	ECCV'22	3 N			N/A	1	Enc-Dec	LG, SEG, SC	❶	
3	UltraLiDAR [397]	CVPR'23	S K P			P	1	Enc-Dec	LG, DET, SC	❶	
4	DUSty v2 [270]	WACV'23	K			L	1	GAN	LG, SEG	❶	
5	LiDARGRIT [116]	arXiv'24	3 K			N/A	1	Enc-Dec, AR	LG	❶	
6	BEVWorld [457]	arXiv'24	C N			T	6	LDM	L4G, DET	❶❷	
7	SDS [87]	arXiv'24	3			P	1	Enc-Dec	LG, SC	❶	
8	HoloDrive [386]	arXiv'24	N			3 T	8	LDM	L4G	❶❸	
9	LOGen [172]	arXiv'24	N			3	N/A	DiT	LG	❶	
10	ViDAR [430]	CVPR'24	N			N/A	6	Enc-Dec	L4G, DET, OP, E2E	❶❷	
11	LiDiff [277]	CVPR'24	3 S			P	1	Enc-Dec	LG, SC	❶	
12	LiDM [300]	CVPR'24	3 N S			S R T	1	LDM	LG	❶	
13	RangeLDM [130]	ECCV'24	3 N			N/A	1	LDM	LG, SC	❶	
14	Text2LiDAR [383]	ECCV'24	3 N			T	1	Enc-Dec	LG, SC	❶	
15	Copilot4D [451]	ICLR'24	A K N			T	6	LDM, AR	L4G	❶❷	
16	R2DM [269]	ICRA'24	3 K			N/A	1	Enc-Dec	LG, SC	❶	
17	3DiSS [278]	arXiv'25	S 3			N/A	1	LDM	LG, SEG	❶	
18	Distill-DPO [460]	arXiv'25	S			P	1	Enc-Dec	LG, SC	❶	
19	DriveX [317]	arXiv'25	N N			T	6	Enc-Dec	L4G, OP, E2E	❶❷	
20	LiSTAR [222]	arXiv'25	N			O	6	AR	L4G	❶❸	
21	LiDARDraft [372]	arXiv'25	3 N S			T R S	1	Enc-Dec	LG, SEG	❶	
22	OLiDM [411]	AAAI'25	3 N			T 3 O	1	Enc-Dec	LG, DET, SC	❶	
23	WeatherGen [384]	CVPR'25	3 S			P	1	Enc-Dec	LG, DET	❶	
24	SuperPC [79]	CVPR'25	3			R P	1	Enc-Dec	LG, SC	❶	
25	OpenDWM [312]	CVPR'25	A 3 N W			3 H	N/A	VQ-VAE	LG, L4G	❶❸	
26	HERMES [485]	ICCV'25	O N			T T	N/A	AR, LLM	L4G, E2E	❶❷	
27	X-Drive [396]	ICLR'25	N			3 T	1	LDM	LG, DET	❶	
28	LidarDM [500]	ICRA'25	3 W			H	N/A	LDM	L4G, DET	❶❸	
29	LiDAR-EDIT [124]	ICRA'25	N			P	1	Enc-Dec	LG, SC, DET	❶	
30	R2Flow [271]	ICRA'25	3 N			N/A	1	DiT	LG	❶	
31	DiffSSC [37]	IROS'25	S 3			P S	1	Enc-Dec	LG, SC	❶	
32	LiDPM [252]	IV'25	S			P	1	Enc-Dec	LG, SC	❶	
33	SPIRAL [490]	NeurIPS'25	S N			S	1	Enc-Dec	LG, SEG	❶	
34	OmniLiDAR [228]	arXiv'26	S 3 N S			T	1	Enc-Dec	LG, SEG, DET	❶	
35	La La LiDAR [229]	AAAI'26	N W			O	1	Enc-Dec	LG, SEG, DET, SC	❶	
36	LiDARCrafter [205]	AAAI'26	N			3 O	6	Enc-Dec	L4G	❶❸	
37	DriveLiDAR4D [34]	AAAI'26	3 N			3 H T	20	Enc-Dec	L4G	❶❸	
38	U4D [405]	CVPR'26	N S 3			P	6	Enc-Dec	L4G, SC	❶	
39	Veila [227]	ICRA'26	K S N			R	1	Enc-Dec	LG, SEG	❶	

Table 4 Continued summary of LiDARGen models. Symbols and categories follow the preceding part. Text names identify additional datasets or modalities; a dash denotes a temporal field not standardized in this summary.

#	Model	Venue	Dataset	Input	Output	Condition	Len.	Arch.	Task	Cat.	URL
40	HelloWorld (Driving) [236]	arXiv'26	P			R	-	DiT	LG	❶	

synthesize traffic participants, conditioned on their relative orientation and distance to the sensor.

Multimodal Generation. Several recent methods [386, 457] investigate multimodal generation by synthesizing aligned LiDAR and image data. X-Drive [396] introduces a dual-branch diffusion architecture for jointly generating aligned LiDAR point clouds and multi-view camera images in driving scenarios. Its key innovation is the cross-modality epipolar condition module, which improves consistency between the point cloud and

image modalities. Furthermore, X-Drive [396] supports controllable 3D scene generation conditioned on heterogeneous inputs, including text descriptions, object bounding boxes, and sensor data variants from the images or the LiDAR point clouds.

HelloWorld [236] adopts a sequential alternative to joint sensor denoising: a separate diffusion branch conditions on multi-view RGB latents and decodes LiDAR range and intensity. This makes the image-to-LiDAR interface explicit, but errors in the conditioning images can propagate into the synthesized sensor observations. Sensor fidelity and cross-modal consistency should therefore be assessed alongside the quality of each modality in isolation.

Label Engines for 4D LiDAR. A complementary use of generation is to manufacture supervision rather than sensor data. Progress on 4D LiDAR segmentation is bottlenecked by the cost of assigning temporally consistent labels across sparse sequences, and each new task or domain tends to demand fresh dense annotation. Kim *et al.* [169] bootstrap a 4D LiDAR annotation tool from video foundation models, asking whether usable LiDAR training data can be produced with no human labelling at all. This positions the data engine one step upstream of the generators above: the artefact produced is an annotated corpus, and the relevant evaluation is downstream task performance rather than generative fidelity.

3.3.2 Action Forecasters

Based on past observations, the LiDAR-based world models functioning as action forecasters generate future LiDAR sequences conditioned on given future states.

Temporal Modeling. Copilot4D [451] proposes a scalable approach to building world models, primarily by (1) leveraging a VQ-VAE [345] model to tokenize complex, unstructured point cloud inputs, and (2) recasting the Masked Generative Image Transformer [41] as a discrete diffusion model to enable parallel denoising and decoding. Copilot4D takes as input 1–3 seconds of past LiDAR frames along with future ego actions (poses), and predicts high-quality LiDAR frames for the next 1–3 seconds. ViDAR [430] takes historical camera frames as input and predicts future LiDAR frames as output. This framework further enables pre-training for tasks such as perception, prediction, and planning.

Multi-Modal Action Forecasters. BEVWorld [457] introduces a multi-modal tokenizer to extend the generative capability to both surround-view images and LiDAR point clouds. DriveX [317] supports multi-modal outputs, including point clouds, camera images, and semantic maps. By employing a decoupled latent world modeling strategy that separates world representation learning for spatial modeling from latent future decoding for future state prediction, DriveX effectively simplifies the modeling of complex dynamics in unstructured scenes. HERMES [485] integrates LLMs to generate textual descriptions of future frames in addition to LiDAR, thereby enhancing human–machine interaction.

3.3.3 Autoregressive Simulators

World models functioning as autoregressive simulators aim to generate temporally coherent LiDAR sequences for realistic and interactive simulation. These models serve as a foundation for perception, planning, and decision-making, with a focus on geometric fidelity and temporal consistency. Existing methods can be divided into two types based on their data generation paradigms.

Sequential Autoregressive LiDAR Generation. HoloDrive [386] presents an autoregressive framework for jointly generating multi-view camera images and LiDAR point clouds by introducing a depth prediction branch in the 2D generative model to improve alignment between 2D and 3D representations. More recently, LiDARCrafter [205] extends the layout-based two-stage framework of La La LiDAR [229] to the 4D domain, with an autoregressive LiDAR sequence generator, supporting fine-grained control, long-term temporal coherence, and diverse editing capabilities. Gen-4D-LiDAR [202] generates 4D LiDAR sequences from instructions with object-level control, LiSTAR [222] builds a ray-centric 4D LiDAR sequence world model, DriveLiDAR4D [34] performs sequential and controllable LiDAR scene generation, and U4D [405] introduces an uncertainty-aware 4D LiDAR world model. More recently, LaGen [482] formulates autoregressive LiDAR scene generation for scalable rollouts, while the Deformable-Mamba LiDAR world model GEM [382] leverages state-space sequence modeling to generate future LiDAR scenes efficiently.

Scene-Scale Simulation from Meshes. LidarDM [500] constructs mesh grids from point clouds by removing dynamic objects across multiple frames. It then trains a diffusion model conditioned on the BEV layout, enabling it to generate a mesh world. By incorporating dynamic objects with motion trajectories into this mesh world and performing ray projection through the scene, LidarDM can synthesize long sequential LiDAR point clouds.

Summary & Insights. LiDARGen has progressed from GAN- and range-image-based synthesis to diffusion, latent-diffusion, and flow-matching models spanning data engines, action forecasters, and autoregressive simulators. The dominant challenges are intrinsic to the sensor: the *sparsity and anisotropic angular sampling* of scans, faithful *ray-drop and intensity modeling*, and preserving native scanline structure under BEV or voxel abstractions. Temporal coherence for 4D sequences remains comparatively underexplored, and cross-modal (LiDAR–camera–occupancy) alignment is still brittle. Sensor-faithful representations, weather- and condition-controllable generation, and long-horizon 4D synthesis are the most impactful open directions.

4 Datasets & Evaluations

In this section, we provide a comprehensive evaluation of 3D/4D world modeling across four aspects. [1]**Datasets** (Sec.4.1) introduce widely used benchmarks with multimodal inputs and annotations across video, occupancy, and LiDAR formats. [2]**Metrics and Protocols** (Sec.4.2) define standardized criteria for assessing generation fidelity, forecasting accuracy, planning awareness, reconstruction quality, and downstream performance. [3]**Quantitative Benchmarks** (Sec.4.3) report results of state-of-the-art models under these protocols. [4]**Qualitative Analyses** (Sec.4.4) highlight strengths, limitations, and trade-offs across different modalities.

4.1 Datasets

In this survey, we discuss real, simulated, and augmented datasets that support research in 3D and 4D world modeling. These datasets span urban driving and related settings and provide rich annotations and conditions needed for **VideoGen**, **OccGen**, and **LiDARGen**. An overview of popular datasets and related benchmarks is illustrated in Figure 3. Additionally, Table 5 provides detailed statistics of each collection of the video, occupancy, LiDAR, and other relevant data formats from these mainstream datasets.

Among existing 3D/4D data collections, real-world datasets supply realism and multimodal context with reliable calibration. Recent web-scale corpora trade strict calibration for scale, diversity, and text supervision. Simulators contribute perfect labels, editable layouts, and rare or counterfactual scenarios. Together, these sources form a complementary foundation for training and evaluating controllable and planning-aware world models.

Video-based datasets provide long, coherent video sequences with reliable calibration, ego pose, and synchronized multi-view images. Conditions that aid controllability include action logs, HD maps, and language signals such as captions or driving commands. Real-world datasets, *e.g.*, nuScenes [32] and Waymo Open [328], provide surround-view imagery, accurate poses, and dense perception annotations, making them strong bases for video generation with map- or motion-conditioned control. Planning-aware datasets like NAVSIM [65] and nuPlan [33] pair short scenarios with ego motion, CAN signals, and maps to support policy-grounded video modeling. Web-scale video such as OpenDV-YouTube [418] contributes breadth and language supervision via captions and ego-action tags, trading off precise calibration for scale and diversity. Synthetic platforms like CARLA [78] offer poses and editable layouts for counterfactuals, rare events, and controlled ablations. Procedural simulators such as TerraZero [388] further enable large-scale, zero-demonstration self-play, generating diverse driving situations without human logs.

Occupancy-based datasets need voxelized 3D supervisions in a consistent coordinate frame, with semantic labels and tight alignment to the sensor rig. Conditions that stabilize learning include HD maps, ego trajectories, and either multi-view images or LiDAR to anchor the field over time. In driving settings, ready-to-use *real-world* benchmarks such as OpenOccupancy [364], Occ3D-nuScenes [339], NYUv2 [321], and SSCBench [199] provide standardized voxel grids and protocols for training and evaluation. *Simulated datasets* like CarlaSC [379] offer clean ground truth and full control of layout and motion, which is useful for ablations and stress tests. Semantic extensions like SemanticKITTI [21] couple point-wise labels with occupancy volumes and enable

Table 5 Summary of datasets and benchmarks used for training **VideoGen**, **OccGen**, and **LiDARGen** models.

• **Column Keys:** **#** = Total number of frames; **#** = Total number of occupancy scenes; **#** = Total number of LiDAR scenes; **Freq** = Annotation frequency; Symbol "–" in a cell indicates the information is not provided.

• **Tasked by:** ❶ Video Gen. Models (**VideoGen**, *cf.* Sec. 3.1), ❷ Occupancy Gen. Models (**OccGen**, *cf.* Sec. 3.2), and ❸ LiDAR Gen. Models (**LiDARGen**, *cf.* Sec. 3.3). Kindly refer to Table 1 for the definitions of conditions.

#	Dataset	Venue	# Scene	# (View)	#	#	Freq	Conditions	Tasked by	URL
K	KITTI [102]	CVPR'12	22	15k (×4)	-	15k	10	D 3 F	❶ ❸	
U	NYUv2 [321]	ECCV'12	464	1449 (×1)	1449	-	-	D S	❷	
C	CARLA [78]	CoRL'17	∞	∞	∞	∞	Free	3 T S	❸	
S	SemanticKITTI [21]	ICCV'19	22	-	43k	23k	10	S T	❷ ❸	
N	nuScenes [32]	CVPR'20	1000	1.4M (×6)	40k	400k	2	3 B H T V S	❶ ❷ ❸	
W	Waymo Open [328]	CVPR'20	1150	1M (×5)	-	230k	10	3 B H T V S	❶ ❷ ❸	
S	STF [25]	CVPR'20	-	1.4M (×2)	-	1.4M	0.1	3 T L P	❸	
V	Virtual KITTI 2 [31]	arXiv'20	5	40k (×2)	-	-	10	3 T	❶	
A	Argoverse 2 [378]	NeurIPS'21	1000	2.7M (×9)	-	150k	10	3 T H	❷ ❸	
L	Lyft-Level5 [125]	CoRL'21	170k	282M (×7)	42.5M	42.5M	10	T H	❷	
N	nuPlan [33]	CVPRW'21	-	24M (×6)	-	24M	-	T H	❶	
P	PandaSet [392]	ITSC'22	103	48k (×6)	-	16k	10	3 L P T S	❸	
O	OpenCOOD [403]	ICRA'22	73	11k (×4)	11k	11k	10	3 T	❷	
3	KITTI-360 [208]	TPAMI'22	379	150k (×4)	-	80k	10	3 L P T S H T	❸	
C	CarlaSC [379]	RA-L'22	24	-	43k	43k	10	T S	❷	
R	Robo3D [174]	ICCV'23	2194	-	-	476k	10	3 T S	❸	
O	OpenOccupancy [364]	ICCV'23	850	200k (×6)	34k	34k	2	3 T S	❷	
N	Occ3D-nuScenes [339]	NeurIPS'23	900	240k (×6)	40k	40k	2	3 T S	❷	
Y	OpenDV-YouTube [418]	CVPR'24	2139	60M (×1)	-	-	10	T C	❶	
S	SSCBench [199]	IROS'24	1859	404k (×6)	66k	66k	-	3 T S	❷	
N	NAVSIM [65]	NeurIPS'24	115k	920k (×8)	-	115k	2	3 T H	❸	
D	DrivingDojo [369]	NeurIPS'24	17.8k	1.7M (×1)	-	-	5	T T	❶	
O	OmniDrive [358]	CVPR'25	1000	1.4M (×6)	40k	-	2	3 T H T	❸	
E	EUVS [117]	ICCV'25	345	90k (×8)	-	-	-	T	❶	
P	Pi3DET [201]	ICCV'25	25	51k (×1)	-	51k	10	3 T	❸	
O	Nuplan-Occ [182]	arXiv'25	19k	3.6M (×8)	3.6M	-	10	3 B H T V S	❶ ❷ ❸	
I	OccInteract-85k [207]	arXiv'26	85k	-	85k	-	-	B C S T	❷	
4	4DLidarOpen [297]	arXiv'26	225	140k (×5)	-	28k	10	3 T F L P	❸	
W	WorldLens [203]	CVPR'26	26.8k	-	-	-	-	3 L P T S H T	❶	

joint learning of geometry and semantics. Large-scale collections such as nuPlan-Occ [182] further provide dense semantic-occupancy supervision tailored to occupancy-centric scene generation.

LiDAR-based datasets require raw LiDAR-acquired sweeps with precise extrinsics, per-sweep ego poses, and object-level annotations. Additional 2D and 3D cues, such as HD maps, radar, and camera imagery, enable cross-modal conditioning, while coverage across weather conditions and sensor configurations improves robustness. Representative real-world sources include KITTI [102], nuScenes [32], Waymo Open [328], and Argoverse2 [378]. NAVSIM [65] supplements these with short scenario snippets paired with control signals, supporting downstream planning tasks. For robustness testing, recent benchmarks [25, 174, 201] capture adverse weather, inject systematic corruptions, and cover multiple platforms to assess generalization. Synthetic platforms, such as CARLA [78], offer clean LiDAR simulations, editable environments, and controllable signals. Emerging sensor modalities are also represented by 4DLidarOpen [297], an open 4D FMCW LiDAR dataset that augments geometry with point-wise radial velocity.

Scene-reconstruction & novel-view-synthesis datasets. Beyond driving-centric corpora, large-scale datasets for general scene reconstruction and novel view synthesis (NVS) are increasingly relevant to 3D/4D world modeling, particularly for the *Scene Reconstructors* functional type. RealEstate10K [483] provides tens of thousands of camera trajectories through indoor and outdoor real-estate video with accurate poses, and ACID [215] offers aerial coastal and nature footage for perpetual view generation; both are widely used to learn and evaluate view synthesis. More recently, DL3DV [213] contributes a large-scale, real-world scene dataset spanning diverse indoor and outdoor environments with dense multi-view coverage, supporting the training and benchmarking of *generalizable* reconstruction and NVS models. Relative to the driving datasets above, these emphasize wide-baseline camera motion, scene-scale geometry, and aggressive view extrapolation, making

them complementary sources for training and evaluating geometry-grounded world models. CG-World [35] takes a different, world-state-first view of the same gap, explicitly capturing the joint dynamics of states, actions, events, and observations that any single one of the corpora above captures only partially.

Controlled benchmarks for 3D/4D representations and closed-loop platforms. Two newer resource families narrow specific gaps left by the corpora above. On the representation side, M^{3}ISR [226] provides a multi-modal multi-view benchmark for 3D/4D Gaussian splatting and feed-forward reconstruction, motivated by the observation that existing real-captured multi-view video benchmarks make it hard to isolate the effects of camera geometry, representation size, and temporal update cost from one another. On the platform side, SIREN-Bench [493] couples SUMO traffic logic with CARLA sensing to generate behaviour-level emergency-vehicle interactions with synchronized observations and simulator ground truth, and V2XBench [407] supplies synchronized ego–roadside sensing with closed-loop evaluation for cooperative driving. Neither platform is a generative world model, but both supply the controlled interaction traces and paired ground truth that closed-loop evaluation of one requires, which is why we group them with datasets rather than with the methods of Sec. 3. Recent resources further isolate the geometric and physical factors that a perceptual score can obscure (Table 6). RoboPhys-3D [360] applies the same reconstruction pipelines to generated and reference manipulation videos, so reconstruction error can be considered when interpreting geometric, state, and task-level scores. VehDyn [361] pairs CARLA rendering with CarSim vehicle dynamics and synchronized state traces, separating trajectory agreement, kinematic consistency, physical feasibility, and responses to changed conditions. These controlled protocols complement broad visual benchmarks; their aggregate rankings remain specific to the reconstruction or state-extraction pipeline and tested scenario distribution.

Table 6 Representative recent protocols for geometry-grounded evaluation. The resources isolate different failure modes; their scores are not directly interchangeable.

Resource	Setting	Reference or control	Evaluation focus
RoboPhys-3D [360]	Robotic manipulation videos	Matched reconstruction of generated and reference videos	Geometry, scene state, and task completeness
VehDyn [361]	Driving videos	CARLA–CarSim paired video and vehicle states	Trajectory, kinematics, feasibility, and condition response
CoDrive-Bench [255]	Multi-vehicle video generation	Shared camera trajectories; real and synthetic scenes	Cross-agent geometry, instance consistency, and control
RealCOD-Rigid [38]	Camera and object control	Canonical object meshes and prescribed rigid motion	Motion adherence with identity preservation (IG-IoU)
GeoWM [276]	Direct depth and geometry forecasts	Metric depth on DOMINO; scale-aligned depth or geometry elsewhere	Horizon-specific forecasting; alignment-dependent accuracy
VGGTWorld-VLA [232]	Supplied-action driving forecasts	VGGT-derived future geometry; ego trajectory and driving intent	Short-horizon teacher agreement, rather than measured dense geometry or policy success
World Observer [61]	Real and CARLA out-of-view sequences	Estimated segmentation and depth under panoramic conditioning	Exit/re-entry and 3D displacement within a single generation chunk
OccStress [473]	4D occupancy forecasting chain	Sensor or occupancy-state perturbations; source-conditioned clean references	Robustness to corruption placement in temporal history, rather than closed-loop safety

4.2 Evaluation Metrics & Protocols

Standardized evaluations lay the foundation for the development of generation models. However, existing literature has overlooked the importance of establishing a systematic protocol for evaluations in 3D and 4D.

Here, we organize evaluation metrics for world models into five perspectives:

- [1]**Generation Quality** (Sec. 4.2.1) assesses the realism, coherence, and controllability of synthesized outputs.
- [2]**Forecasting Quality** (Sec. 4.2.2) evaluates future predictions given partial observations.
- [3]**Planning-Centric Quality** (Sec. 4.2.3) metrics measure safety and rule compliance in planning.
- [4]**Reconstruction-Centric Quality** (Sec. 4.2.4) examines the ability of generation models to reproduce or simulate novel views.
- [5]**Downstream Evaluation** (Sec. 4.2.5) tasks test how world models support tasks like detection, segmentation, and reasoning.

A comprehensive summary of evaluation metrics is provided in Table 15. Together, these metrics cover both perceptual fidelity and utility in embodied decision-making and beyond.

4.2.1 Generation Quality

Generation quality focuses on whether a world model can produce realistic and coherent outputs given a prompt or condition. This involves four dimensions: fidelity, consistency, controllability, and human preference. Dedicated suites such as DrivingGen [487] consolidate these dimensions into a comprehensive benchmark for generative video world models in autonomous driving.

Fidelity evaluates how closely a generator matches the real data distribution and is typically divided into two families. *Perceptual metrics* project samples into a feature space learned from human-labeled data, where distances align with human judgments of realism. The Fréchet family [122, 172, 230, 269, 300, 320, 344] encodes samples, fits Gaussians to real and generated features, and reports the Fréchet distance. Some variants differ by modality and encoder, while semantic versions [490] add labels to align categories. Other representative metrics include Inception Score [308], which uses Inception logits to reward confident and diverse predictions without real references. *Statistical metrics* operate directly on geometry or density. They ask whether the generated set covers the real set, stays within it, and matches the low-level structure. Some metrics [24, 172] target the fidelity–coverage trade-off, probing set overlap by measuring whether generated samples stay on the real manifold while sufficiently covering it, while other metrics [116, 499] quantify distributional discrepancy in geometry or density via different distance metrics.

Consistency evaluates whether a world model produces coherent outputs across space, time, and semantics. *Spatial Consistency* scores geometric alignment. Some [330, 368] quantify multi-view agreement by matching keypoints in overlapping regions, while others evaluate alignment by projecting the 3D outputs and comparing them with monocular depth estimates [396]. *Temporal Consistency* is measured by cosine similarity [154] between adjacent-frame embeddings from foundation models [282, 299], and *Subject Consistency* [144] tracks identity persistence by comparing subject-region features [282] across frames.

Table 7 Benchmarking **VideoGen** models on the **Perceptual Fidelity** of generation quality evaluations. The reported metrics are FID and FVD scores on the official nuScenes [32] validation set. All metrics are the lower the better (↓).

Method	Resolution	Freq	FID ↓	FVD ↓
Single-View Video Generation				
DriveDreamer [365]	128×192	2 Hz	14.90	340.80
GenAD [418]	256×448	2 Hz	15.40	184.00
ProphetDWM [362]	256×448	2 Hz	6.90	190.50
Epona [449]	512×1024	5 Hz	7.50	82.80
MaskGWM [274]	288×512	10 Hz	4.00	59.40
LongDWM [363]	480×720	10 Hz	12.30	102.90
DriVerse [194]	480×832	10 Hz	18.20	95.20
InfinityDrive [114]	576×1024	10 Hz	10.93	70.06
GEM [118]	576×1024	10 Hz	10.50	158.50
Vista [98]	576×1024	10 Hz	6.90	89.40
UniFuture [206]	320×576	12 Hz	11.80	99.90
MiLA [351]	360×640	12 Hz	8.90	89.30
GeoDrive [43]	480×720	12 Hz	4.10	61.60
STAGE [353]	512×768	12 Hz	11.04	242.79
Doe-1 [471]	384×672	-	15.90	-
Multi-View Video Generation				
Drive-WM [368]	192×384	2 Hz	15.80	122.70
WoVoGen [239]	256×448	2 Hz	27.60	417.70
Panacea [375]	256×512	2 Hz	16.96	139.00
SubjectDrive [136]	256×512	2 Hz	15.98	124.00
Glad [393]	256×512	2 Hz	11.18	188.00
SynthOcc [190]	448×800	2 Hz	14.75	-
CogDriving [237]	480×720	2 Hz	15.30	37.80
DrivingDiffusion [195]	512×512	2 Hz	15.83	332.00
Delphi [245]	512×512	2 Hz	15.08	113.50
MaskGWM [274]	288×512	10 Hz	8.90	65.40
DriveScape [381]	576×1024	10 Hz	8.34	76.39
MagicDrive3D [95]	224×400	12 Hz	20.67	164.72
MagicDrive [97]	224×400	12 Hz	16.20	218.12
DreamForge [253]	224×400	12 Hz	14.61	209.90
DrivePhysica [432]	256×448	12 Hz	3.96	38.06
UniScene [181]	256×512	12 Hz	6.45	71.94
MiLA [351]	360×640	12 Hz	4.90	36.30
CoGen [151]	360×640	12 Hz	10.15	68.43
DiST-4D [113]	424×800	12 Hz	6.83	22.67
DiVE [154]	480×854	12 Hz	-	94.60
DrivingSphere [409]	480×1080	12 Hz	-	103.40
DriveDreamer-2 [464]	512×512	12 Hz	11.20	55.70
NoiseController [73]	512×1024	12 Hz	13.72	87.23
MagicDrive-V2 [96]	848×1600	12 Hz	20.91	94.84
BEVWorld [457]	-	12 Hz	19.00	154.00
UniMLVG [48]	-	12 Hz	5.80	36.10
DualDiff [188]	224×400	-	10.99	160.00
BEVGen [330]	224×400	-	24.54	-
PerLDiff [447]	256×708	-	13.36	-
HoloDrive [386]	-	-	13.60	103.00
BEVControl [420]	-	-	24.85	-

Controllability measures how well a model adheres to user-specified inputs, with metric design tailored to the conditioning modality. When the condition is reference frames, CLIP Similarity [245, 418] averages cosine similarity between CLIP embeddings of generated and reference frames to gauge semantic alignment. Beyond this, layout and object-level control is typically scored by agreement with detectors or segmentors on boxes and masks [488], scene-graph control by count errors and set overlap [231], and camera-pose control by trajectory rotation and translation errors [160],

Table 8 Benchmarking **VideoGen** models on the **Downstream Evaluation** tasks. The reported metrics are mAP and NDS for **3D Object Detection**, mIoU (Lanes, Drivable, Divider) for **BEV Map Segmentation**, L2 and Collision Rates at timestamps 1s, 2s, and 3s for **Open-Loop Planning**, and PDMS (P) and ADS (A) scores [423] for **Closed-Loop Planning**. All results are computed using the UniAD [135] implementation and checkpoints on official nuScenes [32] validation set.

Method	3D Det ↑		BEV Seg mIoU (%) ↑			Open-Loop Planning ↓						Closed-Loop Planning ↑			
	mAP	NDS	Lane	Dri	Div	L2@1s	L2@2s	L2@3s	CR@1s	CR@2s	CR@3s	P@SG	A@SG	P@BOS	A@BOS
Baseline [135]	37.98	49.85	31.31	69.14	25.93	0.51	0.98	1.65	0.10	0.15	0.61	-	-	-	-
MagicDrive [97]	12.92	28.36	21.95	51.46	17.10	0.57	1.14	1.95	0.10	0.25	0.70	-	-	-	-
Panacea [375]	13.72	27.73	18.23	52.37	17.21	0.58	1.14	1.95	-	-	-	-	-	-	-
DiST-4D [113]	15.63	32.44	26.80	60.32	21.69	0.56	1.11	1.91	-	-	-	-	-	-	-
DriveArena [423]	16.06	30.03	26.14	59.37	20.79	0.56	1.10	1.89	0.02	0.18	0.53	0.76	0.13	0.50	0.045
DreamForge [253]	16.63	30.57	26.16	58.98	20.22	0.55	1.08	1.85	0.08	0.27	0.81	0.81	0.12	0.74	0.076
DrivingSphere [409]	21.45	34.16	57.99	62.87	22.29	0.54	1.10	1.76	-	-	-	-	-	-	-

Human Preference captures subjective qualities like realism and plausibility that automated scores may miss. Studies typically adopt either two-alternative forced choice [363] or mean opinion score [231] setups, involving both experts and lay users to provide human evaluation on world models.

Holistic Benchmarking. The four dimensions above are most often reported in isolation, making it difficult to determine whether visual quality is accompanied by geometric coherence and useful behavior. WorldLens [203] evaluates driving world models along five complementary aspects—generation, reconstruction, action-following, downstream tasks, and human preference—through 24 standardized dimensions. Its reconstruction protocol fits generated videos with 4D Gaussian representations and assesses geometric and novel-view fidelity. Action-following separates open-loop displacement and adherence from closed-loop route completion and adherence, while downstream evaluation applies frozen perception models to map segmentation, 3D detection, tracking, and occupancy prediction. These tests expose different failure modes along the generation-to-driving pipeline, so improvements in image quality alone do not establish geometric accuracy or downstream utility.

WorldLens also pairs *WorldLens-26K*, a 26,808-entry human-annotated preference dataset containing scores and textual rationales, with *WorldLens-Agent*, a vision-language evaluator trained on these annotations for explainable assessment [203]. Its results reveal that no evaluated approach dominates all aspects: perceptual realism, geometric fidelity, and behavioral quality remain distinct capabilities. The learned evaluator makes human-aligned assessment more scalable, while explicit geometry, task, and closed-loop protocols retain separate roles in testing the generated environment. This motivates the behavior- and physics-oriented criteria in Sec. 4.2.3 and Sec. 4.2.6.

Orbia [235] complements this driving-oriented coverage with controlled exploration-and-revisit tests of whether generated videos maintain a consistent 3D world. Across 720 evaluation cases, six trajectory templates, and 27 models, it distinguishes *input preservation* of initially observed regions, *generated-content persistence* of regions first revealed during exploration, and *3D self-consistency* across generated views. Registered RGB-D references support preservation checks on observed surfaces; newly generated regions are instead compared across their first appearance and subsequent revisits. This distinction avoids treating unobserved content as if it had a ground-truth appearance.

For self-consistency, Orbia combines local geometric agreement with reconstruction from context frames and rendering of held-out generated views [235]. Geometry is evaluated using recovered camera poses, with adherence to the requested controls scored separately. Its short and minute-long revisit protocols expose forgetting and geometric drift that can remain hidden in visually plausible clips or adjacent-frame comparisons. Together, WorldLens and Orbia show why evaluation should connect perceptual quality to geometry, memory, control, and task behavior, while keeping the reference coverage, camera estimation, and model-interface adaptations explicit when interpreting results.

4.2.2 Forecasting Quality

Forecasting quality extends beyond unconditional generation by evaluating how well the model predicts the future evolution of a scene given partial observations. Here, forecasting quality is evaluated in spatial and temporal domains.

Table 9 Benchmarking **VideoGen** models on the **Downstream Evaluation** tasks. The reported metrics are mAP and NDS for **3D Object Detection** (*w/* BEVFusion [233] and StreamPETR [357]) and Road-wise mIoU scores (RmIoU) and Vehicle-wise mIoU scores (VmIoU) for **BEV Map Segmentation** (*w/* CVT [475]). The results are on the official nuScenes [32] validation set. All metrics are the higher the better (↑).

Method	BEVFusion		StreamPETR		CVT	
	mAP	NDS	mAP	NDS	RmIoU	VmIoU
Baseline	35.54	41.21	34.50	46.90	73.67	34.82
BEVControl [420]	-	-	-	-	60.80	26.80
BEVGen [330]	-	-	-	-	50.20	5.89
Panacea [375]	-	-	22.50	36.10	-	-
DrivingDiffusion [195]	-	-	-	-	63.20	31.60
SimGen [488]	-	-	-	-	62.90	31.20
CogDriving [237]	-	-	-	-	65.70	32.10
UniMLVG [48]	-	-	-	-	70.81	29.12
DrivePhysica [432]	-	-	35.50	43.67	-	-
SubjectDrive [136]	-	-	28.00	41.10	-	-
Glad [393]	-	-	27.10	40.80	-	-
DriveScape [381]	-	36.50	-	-	64.43	28.86
MagicDrive [97]	12.30	23.32	-	-	61.05	27.01
DreamForge [253]	13.01	22.16	26.00	41.10	65.27	28.36
DualDiff [188]	13.99	24.98	-	-	62.75	30.22
PerLDiff [447]	15.24	24.05	-	-	61.26	27.13
MagicDrive-V2 [96]	17.65	-	-	-	59.79	32.73
NoiseController [73]	20.93	27.96	-	-	64.85	27.32
DrivingSphere [409]	22.71	31.79	-	-	-	-

Spatial Predictive Accuracy in forecasting measures how well predictions match the actual future in the spatial domain. For frames and videos, FID, FVD, and frame-level L1/L2 errors remain standard. IoU evaluates occupancy forecasts [246] at multiple horizons to separate near- and long-range correctness. Point-cloud forecasts [451] are evaluated by comparing the predicted and ground-truth sweeps in 3D space, using Chamfer distance for geometric overlap and depth-wise errors to quantify per-ray distance accuracy.

Direct geometry forecasting also requires protocol-aware comparison. GeoWM [276] predicts future depth at a requested horizon from geometric history and a forecast camera viewpoint, avoiding recursive rollout. Its evaluation separates raw metric depth on DOMINO from scale-aligned depth and geometry on other datasets; these settings should not be pooled as equivalent measures of absolute geometric accuracy. VGGTWorld-VLA [232] instead conditions future geometry tokens on driving semantics and a supplied ego trajectory. Its short-horizon NAVSIM evaluation compares decoded geometry with VGGT-derived future references, so the reported agreement characterizes action-conditioned prediction against a geometric teacher rather than measured dense ground truth or closed-loop driving performance.

Temporal Predictive Accuracy in 4D forecasting assesses whether predictions remain temporally coherent, especially without full supervision [367]. Typical examples are Key Object Dimension Probability [367], which penalizes unlikely object sizes using category-specific priors, and Temporal Background Environment Consistency [367], which tracks static voxels under ego-motion to verify scene rigidity.

Clean occupancy forecasting scores also leave the perception–forecasting chain under-specified. OccStress [473] separates sensor corruptions propagated through upstream occupancy estimators from controlled perturbations of the occupancy states themselves, and varies whether errors affect the current state, recent history, or earlier context. Source-conditioned clean references help distinguish perception quality from a forecaster's sensitivity to temporal errors. These diagnostics complement clean IoU and mIoU by testing whether history suppresses or propagates faulty evidence; they do not replace closed-loop evaluation or establish deployment safety.

4.2.3 Planning-Centric Quality

Planning-centric metrics assess whether the model's outputs result in safe, efficient, and rule-compliant decisions, and its evaluation falls into open-loop and closed-loop.

Open-Loop Planning assessment evaluates predictions that do not influence future inputs. nuPlan [33] compares predictions to expert demonstrations using waypoints and heading error, and a horizon-dependent Miss Rate, which thresholds trajectory and heading errors into bounded scores. To approximate behavioral quality without full interaction, NAVSIM [39, 65] introduces short non-reactive rollouts and aggregate safety, drivable-area compliance, progress, and comfort into a single policy score, using gating and weighted averaging to align with closed-loop outcomes.

Closed-Loop Planning evaluation executes the policy in an interactive simulator and scores observed behavior. CARLA [78] reports route or goal completion and infraction distance statistics for opposite-lane driving, sidewalk incursions, and collisions with other agents. nuPlan [33] provides a broader suite of closed-loop checks, including no at-fault collisions, drivable-area and direction compliance, time-to-collision bounds, speed-limit compliance, progress along route, capturing both traffic legality and human-likeness. Recent benchmarks push this paradigm further: World-in-World [445] scores world models by closed-loop task success rather than visual quality, DriveE2E [436] embeds digital twins of real intersections into CARLA for

real-to-sim closed-loop testing, nuPlan-R [288] replaces log-replay traffic with reactive diffusion-based agents, and ReactSim-Bench [459] targets reactive behavior simulation and counterfactual reactivity.

4.2.4 Reconstruction-Centric Quality

Reconstruction-centric neural simulators aim to reproject the past into interactive sensor views or novel viewpoints.

Photometric Fidelity captures low-level rendering quality when ground-truth images under known viewpoints are available. Following standard practices in neural rendering, metrics such as PSNR [146], SSIM [370], and LPIPS [454] remain foundational. PSNR quantifies pixel-level accuracy, SSIM evaluates structural consistency in luminance and texture, while LPIPS measures perceptual similarity in deep feature space aligned with human visual preferences.

Table 10 Benchmarking **OccGen** models on **Reconstruction Quality**. The reported metrics are mIoU (%) for **Semantic Occupancy Reconstruction** and IoU (%) for **Occupancy Reconstruction**. All results are on the official nuScenes [32] validation set. Both metrics are the higher the better (↑).

Method	Type	Resolution	mIoU ↑	IoU ↑
OccSora [355]	VQVAE	$(\frac{T}{8}, 25, 25, 512)$	27.40	37.00
OccLLaMA [373]	VQVAE	$(50, 50, 128)$	65.93	57.66
OccWorld [469]	VQVAE	$(50, 50, 128)$	66.38	62.29
UrbanDiff [448]	VQVAE	$(50, 50, 2048)$	80.00	98.80
I^2World [209]	VQVAE	$(50, 50, 128)$	81.22	68.30
Occ-LLM [404]	VAE	$(50, 50, 64)$	71.08	62.74
UniScene [181]	VAE	$(50, 50, 8)$	72.90	64.10
DOME [109]	VAE	$(25, 25, 64)$	83.08	77.25
UniScene [181]	VAE	$(100, 100, 8)$	92.10	87.00
T^3Former [398]	Triplane-VAE	$(100, 100, 16, 8)$	85.50	72.07
X-Scene [425]	Triplane-VAE	$(100, 100, 16, 8)$	92.40	85.60

View Changing Consistency evaluates the spatiotemporal plausibility of novel or counterfactual viewpoints where ground truth is unavailable [273, 463]. In such settings, photometric comparison is insufficient. Metrics like Novel Trajectory Agent IoU [463] assess whether foreground agents maintain geometrically plausible behavior, offering targeted signals for validating realism in 4D interactive simulations.

4.2.5 Downstream Evaluation

While the above evaluations assess a world model in isolation, downstream evaluations measure its utility when integrated into end-to-end perception and decision-making systems. Tasks span *object detection* (mAP [212], nuScenes Detection Score [32]), *multi-object tracking* (MOTA, MOTP [22]), *semantic and BEV map segmentation* (mIoU), *3D occupancy prediction and scene completion* (voxel-level IoU, Voxelized Panoptic Quality). In language-grounded settings such as *visual question answering*, models like OccLLaMA [373] report exact-match Top-1 accuracy across question types and difficulty levels. These evaluations reflect how well a learned world model supports downstream reasoning, representation, and control tasks effectively.

Distributional, Memory, and Failure-Mode Probes. A parallel line of evaluation questions the unit of assessment itself rather than adding another quality score. PAWBench [294] observes that existing protocols score the plausibility of individual videos, which cannot reveal whether repeated generations recover the *distribution* of admissible futures; it formalizes probabilistic alignment as a distributional criterion, a sharper requirement than any single-rollout metric for a model claimed to be a world model. R2M-Bench [108] identifies a confound specific to long-horizon interactive rollouts: high similarity between a first visit and a return to the same place does not establish that the model remembered anything, because the intervening rollout may simply have changed very little, so absolute revisit scores are contaminated by rendering stability and failed motion; it therefore scores revisit memory *relatively*. Complementing both, Shi *et al.* [319] search for natural-input failures rather than averaging over a fixed test set, on the argument that a world model used as an internal simulator propagates its catastrophic predictions into everything downstream. These probes are methodologically continuous with the physics- and causality-oriented criteria developed next.

4.2.6 Toward Physics- and Causal-Consistency Evaluation

We stress that the perceptual and distributional metrics above (*e.g.*, FID, FVD, mIoU, and Chamfer distance) primarily quantify statistical alignment and visual realism; they are largely blind to whether a generated 3D/4D world is *physically compliant* and *causally correct*. A scene may attain excellent FID/FVD yet violate rigid-body dynamics, permit object inter-penetration, or respond inconsistently to ego actions. We therefore advocate complementing them with evaluation rooted in physical and causal consistency: (i) geometric/collision and kinematic-plausibility checks; (ii) multi-view and temporal geometric-consistency

tests; (iii) action-conditioned counterfactual probes that verify the response to interventions (*e.g.*, that braking actually slows the scene); and (iv) closed-loop, task-grounded outcomes as a proxy for actionable correctness. Establishing such standardized, physics-aware benchmarks is, in our view, among the most pressing needs for the field (*cf.* Sec. 6). Early steps in this direction include admissibility certification for world-model simulators [281], causal-consistency probes that expose how video diffusion degrades as causal chains lengthen [42], and VLM-guided semantic auditing of occupancy world-model rollouts [390]. Recent benchmarks operationalize exactly this distinction: Apple-π [433] verifies whether a video generator arrives at a physically plausible outcome via the correct intermediate reasoning, not merely whether the output looks law-abiding; Thinking in Video [456] redefines video generation itself as a reasoning paradigm and probes whether generators can genuinely reason about real-world dynamics rather than pattern-match training footage; WorldExam [428] separates apparent appearance from inherent reactivity, testing whether a model infers how the world *should* respond rather than merely how it looks; and GAUGE [359] grounds physical-fidelity evaluation in explicit measurement, identifying which physical principle or parameter a rollout violates rather than reporting a single aggregate realism score.

4.3 Quantitative Experiments & Analyses

In this section, we quantitatively evaluate world modeling approaches through [1]**VideoGen Benchmarks** (Sec.4.3.1), [2]**OccGen Benchmarks** (Sec.4.3.2), and [3]**LiDARGen Benchmarks** (Sec. 4.3.3). Models are assessed on standardized datasets using fidelity, consistency, and forecasting metrics, along with downstream perception and planning tasks. These evaluations reveal both the progress and limitations of current methods, highlighting key trade-offs between realism, geometric accuracy, temporal stability, and controllability.

Table 11 Benchmarking **OccGen** models on **4D Occupancy Forecasting Quality**. The reported metrics are mIoU (%) for **Semantic Occupancy Reconstruction** and IoU (%) for **Occupancy Reconstruction**, respectively, at timestamps 1s, 2s, and 3s. All results are on the official nuScenes [32] validation set. Both metrics are the higher the better (↑).

Method	mIoU (%) ↑			IoU (%) ↑		
	1s	2s	3s	1s	2s	3s
GaussianAD [472]	6.29	5.36	4.58	14.13	14.09	14.04
PreWorld [193]	12.27	9.24	7.15	23.62	21.62	19.63
Occ-LLM [404]	24.02	21.65	17.29	36.65	32.14	28.77
OccLLaMA [373]	25.05	19.49	15.26	34.56	28.53	24.41
OccWorld [469]	25.78	15.14	10.51	34.63	25.07	20.18
RenderWorld [416]	28.69	18.89	14.83	37.74	28.41	24.08
COME [318]	30.57	19.91	13.38	36.96	28.26	21.86
DFIT-OccWorld [444]	31.68	21.29	15.18	40.28	31.24	25.29
DOME [109]	35.11	25.89	20.29	43.99	35.36	29.74
UniScene [181]	35.37	29.59	25.08	38.34	32.70	29.09
T^3Former [398]	46.32	33.23	28.73	77.00	75.89	76.32
I^2World [209]	47.62	38.58	32.98	54.29	49.43	45.69

A note on evaluation protocols. Unless otherwise stated, the quantitative results compiled in the following tables are **aggregated from the respective original publications** (and official leaderboards where available) rather than re-evaluated by us under a single unified codebase. We group only comparable settings – matching datasets, splits, resolutions, and metric definitions wherever reported – but residual discrepancies across papers (*e.g.*, differing FVD feature extractors, frame counts and sampling schemes, point-cloud ranges, or voxel resolutions) may still affect strict cross-method comparability. The reported numbers should therefore be read as indicative of general trends rather than the outcome of a controlled, like-for-like re-benchmark, and we accordingly flag a unified, open-source evaluation codebase for 3D/4D world models as an important direction for the community.

4.3.1 Benchmarking Video Generation Models

Generation Fidelity. Table 7 reports FID and FVD results on the nuScenes validation set for both single-view and multi-view vision-based world models. Early baselines such as GenAD [418] and DriveDreamer [365] operate at relatively low resolutions and frame rates, achieving modest performance (FID ∼15, FVD 180–340). Later single-view models improve visual quality. Vista [98] and InfinityDrive [114] leverage higher resolutions and frame rates, reducing FVD below 100. Recent works like MaskGWM [274] and GeoDrive [43] set new state-of-the-art, reaching FID around 4–5 and FVD near 60. In the multi-view setting, early BEV-based approaches (BEVControl [420], BEVGen [330]) yield high FID (>20). Subsequent models, including DriveWM [368], Panacea [375], and MagicDrive [97] reduce errors but struggle with temporal stability (FVD >120). Strong improvements come from models emphasizing geometric consistency and spatio-temporal alignment. UniScene [181], DriveScape [381], and DiST-4D [113] achieve the best balance, with FVD scores below 80 and DiST-4D [113] reaching as low as 22.67.

The comparison suggests resolution and frame rate strongly influence generation fidelity. Besides, explicit multi-view modeling is challenging; although many methods reduce FID, temporal coherence remains difficult, highlighting the importance of structured 4D representations. Finally, methods combining geometry-aware priors with temporal reasoning, such as DiST-4D and UniScene, demonstrate that enforcing spatial structure and temporal consistency jointly is crucial for scalable autonomous driving video generation.

Table 12 Benchmarking **OccGen** models on **Motion Planning Quality**. The reported metrics are L2 Error Rate (in meters) and Collision Rate (%), respectively, at timestamps 1s, 2s, and 3s. All results are on the official nuScenes [32] validation set. Both metrics are the lower the better (↓).

Method	L2 Error (m) ↓			Collision Rate (%) ↓		
	1s	2s	3s	1s	2s	3s
ST-P3 [131]	1.33	2.11	2.90	0.23	0.62	1.27
OccNet [341]	1.29	2.13	2.99	0.21	0.59	1.37
FSF-Net [112]	0.54	1.09	-	0.01	0.01	-
UniAD [135]	0.48	0.96	1.65	0.05	0.17	0.71
OccWorld [469]	0.43	1.08	1.99	0.07	0.38	1.35
PreWorld [193]	0.41	1.16	2.32	0.50	0.88	2.42
GaussianAD [472]	0.40	0.64	0.88	0.09	0.38	0.81
DFIT-OccWorld [444]	0.38	0.96	1.75	0.07	0.39	0.90
Occ-LLaMA [373]	0.37	1.02	2.03	0.04	0.24	1.20
GenAD [418]	0.36	0.83	1.55	0.06	0.23	1.00
RenderWorld [416]	0.35	0.91	1.84	0.05	0.40	1.39
T^3Former [398]	0.32	0.91	1.76	0.08	0.32	0.51
Drive-OccWorld [426]	0.32	0.75	1.49	0.05	0.17	0.64
Occ-LLM [404]	0.12	0.24	0.49	-	-	-

Downstream Evaluations. Table 9 and Table 8 evaluate downstream perception and planning on generated scenes. Early generative baselines (BEVControl [420], BEVGen [330]) provide limited perception benefits, especially in vehicle segmentation ($<$ 27% mIoU). More advanced methods such as MagicDrive [97] and DreamForge [253] improve both detection (up to 26 mAP on StreamPETR [357]) and segmentation ($>$ 61% road mIoU), while DrivePhysica [432] and Glad [393] further push detection accuracy (35.5 mAP, 43.7 NDS). For segmentation, UniMLVG [48] and CogDriving [237] achieve the highest fidelity (70.8% road, 32.1% vehicle mIoU). Beyond perception, planning performance highlights the persistent gap between synthetic and real data. While real nuScenes provides the upper bound (37.9 mAP, 49.9 NDS, 1.05 Avg L2), generative methods lag significantly in detection and planning accuracy. Nevertheless, world models like DriveArena [423] and DreamForge demonstrate reduced planning errors and collision rates, enabling preliminary closed-loop driving with non-trivial success rates (*e.g.*, 0.81 PDMS). DrivingSphere [409] achieves the strongest drivable area segmentation (>58% mIoU), while DiST-4D [113] balances detection and segmentation performance but lacks closed-loop validation.

Overall, the results show that photorealistic generation alone is insufficient to improve downstream tasks; explicit modeling of geometry, temporal consistency, and motion dynamics is crucial. Models that incorporate such priors not only enhance detection and segmentation but also support safer planning by reducing collisions and trajectory errors. Strong segmentation fidelity further demonstrates the benefit of multi-view and structure-aware models in capturing global layouts, yet the performance gap to real data remains significant, underscoring the challenge of aligning generative fidelity with task-level utility.

4.3.2 Benchmarking Occupancy Generation Models

Occupancy Reconstruction Quality. Table 10 evaluates the reconstruction capability of occupancy world models under VAE-based formulations. Conventional VAEs such as DOME [109] already achieve strong results (83.08% mIoU, 77.25% IoU), outperforming most VQVAEs. While UrbanDiff [448] and I^2World [209] show competitive IoU, other variants like OccSora [355] degrade significantly under coarse temporal–spatial compression. Triplane-based VAEs [180, 398, 425] bring the largest gains, with T^3Former [398] reaching 85.50% mIoU and X-Scene [425] establishing a new state-of-the-art at 92.40% mIoU and 85.60% IoU.

These results underline that latent representation design is decisive for reconstruction fidelity. Triplane factorization enforces geometric consistency and enables finer spatial detail, while simply enlarging latent dimensionality (*e.g.*, UrbanDiff [448] with 2048 channels) yields limited returns. Compact VAEs such as UniScene [181] further show that well-regularized low-dimensional spaces can generalize effectively, whereas aggressive compression (*e.g.*, OccSora [355]) sacrifices accuracy. Overall, effective compression combined with explicit geometric priors is key to scalable and accurate 3D and 4D scene modeling.

4D Occupancy Forecasting Quality. Table 11 presents 4D occupancy forecasting results over the period of 1–3 seconds. Baselines such as OccWorld [469] and OccLLaMA [373] achieve moderate performance (17–20% mIoU), while DOME [109] and UniScene [181] improve temporal stability (27.10% and 31.76% mIoU). More recent models show further progress: I^2World [209] reaches 39.73% mIoU with balanced IoU, and

T^{3}Former [398] excels in spatial coherence with 76.40% IoU. The comparisons reveal three insights. First, naive autoregressive or generative approaches deteriorate rapidly at longer horizons, highlighting the need for structured priors. Second, triplane factorization substantially improves spatial fidelity, as reflected in the performance of T^{3}Former [398]. Third, I^{2}World shows that coupling scalable latent reasoning with temporal modeling yields the best balance across horizons. Accurate 4D forecasting thus requires not only generative power but also structured representations that enforce geometric and temporal consistency.

End-to-End Planning. Table 12 reports the performance of end-to-end planning, measured by trajectory error (L2) and collision rate. Sequence-based planners like ST-P3 [131] perform poorly (2.11 meters in L2 error), while UniAD [135] and GenAD [418] achieve substantial gains, with UniAD+DriveWorld [262] further improving to 0.69 meter in L2 error and 0.19% collisions. Occupancy-based world models such as OccWorld [469] and OccLLaMA [373] reduce errors to around 1.15 meters. Structured refinements (*e.g.*, DFIT-OccWorld [444], RenderWorld [416], and Drive-OccWorld [426]) achieve stronger accuracy and safety, with Drive-OccWorld reaching 0.85 m in L2 error and 0.29% collisions. Notably, GaussianAD [472] and T^{3}Former [398] balance error and safety, while Occ-LLM [404] reports extremely low error (*i.e.*, only 0.28 meter in L2 error).

The results show that integrating occupancy world models into planning pipelines consistently outperforms pure trajectory-based methods. Hybrid designs that refine occupancy priors, such as Drive-OccWorld [426] and DFIT-OccWorld [444], bring joint improvements in accuracy and safety, demonstrating the downstream robustness of generative modeling. Overall, structured occupancy representations form a strong foundation for end-to-end autonomous driving, enabling reliable long-horizon planning in complex scenarios.

Table 13 Benchmarking **LiDARGen** models on the **Perceptual Fidelity** evaluations. The reported metrics are FRD, FPD, JSD, and MMD scores on the official SemanticKITTI [21] dataset. All metrics are the lower the better (↓).

Method	Resolution	FRD↓	FPD↓	JSD↓	MMD↓
LiDARGen [499]	64×1024	681.37	115.17	0.1323	2.19×10^{-3}
LiDM [300]	64×1024	-	108.70	0.0456	2.90×10^{-4}
R2DM [269]	64×1024	192.81	19.29	0.0373	1.60×10^{-4}
Text2LiDAR [383]	64×1024	522.32	11.09	0.0750	4.29×10^{-4}
WeatherGen [384]	64×1024	184.11	11.42	0.0290	3.80×10^{-5}

Table 14 Benchmarking **LiDARGen** models on **4D LiDAR Generation Quality**. The reported metrics are TTCE and CTC. The numbers indicate frame intervals. All results are on nuScenes [32]. Both metrics are the lower the better (↓).

Method	TTCE↓		CTC↓			
	3	4	1	2	3	4
UniScene [181]	2.74	3.69	0.90	1.84	3.64	3.90
OpenDWM [312]	2.68	3.65	1.02	2.02	3.37	5.05
OpenDWM-DiT [312]	2.71	3.66	0.89	1.79	3.06	4.64
LiDARCrafter [205]	2.65	3.56	1.12	2.38	3.02	4.81

4.3.3 Benchmarking LiDAR Generation Models

Generation Fidelity. Table 13 reports the performance of recent LiDAR scene generation methods on SemanticKITTI [21] using four fidelity metrics (FRD, FPD, JSD, and MMD). Earlier methods such as LiDARGen [499] and LiDM [300] exhibit relatively large distributional discrepancies, as reflected by high FRD and FPD scores. In contrast, more recent approaches, including R2DM [269], Text2LiDAR [383], and WeatherGen [384], achieve substantially better results across most metrics, indicating a closer alignment between generated and real LiDAR distributions.

The results reveal a clear progression in LiDAR generation quality. Among evaluated methods, WeatherGen [384] achieves the best performance across all metrics by employing Mamba [106] as its backbone. Interestingly, Text2LiDAR [383], despite its strong conditioning on textual input, produces higher FRD, suggesting that aligning with semantic prompts may compromise geometric fidelity. These findings underscore the importance of balancing semantic controllability with distributional realism in future LiDAR scene generation research.

4D LiDAR Generation Quality. Table 14 benchmarks recent LiDAR-based 4D scene generation methods on temporal coherence, using TTCE (Temporal Transformation Consistency Error) and CTC (Chamfer Temporal Consistency) as evaluation metrics. Unlike video generation, which has been extensively studied with standardized benchmarks, temporal LiDAR generation remains relatively underexplored, and current metrics mainly focus on explicit geometric alignment across frames. The results reveal several observations. First, end-to-end autoregressive methods such as UniScene [181] and OpenDWM-DiT [312] demonstrate clear advantages in maintaining short-horizon geometric consistency, as reflected in lower TTCE and CTC at 1–2



Figure 7 Qualitative comparisons of state-of-the-art **VideoGen** models on the nuScenes [32] dataset. From top to bottom rows: Reference (from the dataset), MagicDrive [97], DreamForge [253], DriveDreamer-2 [464], and OpenDWM [312].

frame intervals. However, their fixed-length generation limits broader applicability, as error accumulation grows at longer horizons. Second, incorporating strong vector quantization modules [312] facilitates better condition embedding and fine-grained reconstruction, leading to improved temporal stability. Third, modality choices introduce inherent trade-offs: BEV-based generation offers smoother temporal continuity but sacrifices fidelity to the raw point cloud pattern, while range-based [205] generation better preserves LiDAR-specific sensing characteristics but requires careful design to embed conditions and sustain long-term consistency.

4.4 Qualitative Experiments & Analyses

In this section, we qualitatively evaluate the 3D and 4D generation approaches through [1]**VideoGen Visualizations** (Sec.4.4.1), [2]**OccGen Visualizations** (Sec.4.4.2), and [3]**LiDARGen Visualizations** (Sec. 4.4.3). These evaluations highlight the strengths, limitations, and trade-offs of current methods, informing future advances in realism, consistency, and generalization.

4.4.1 Qualitative Analyses of VideoGen Models

Visual Realism. Figure 7 and Figure 8 compare recent video generation world models, including MagicDrive [97], DreamForge [253], DriveDreamer-2 [464], and OpenDWM [312]. The generated scenes capture overall layouts and semantics close to real-world distributions, but fine-grained details often suffer from pixel misalignment, blurred textures, and structural discontinuities. Among the methods, OpenDWM [312] achieves the most realistic, consistent, and controllable results, owing to its training on diverse datasets (OpenDV [418], nuScenes [32], and Waymo Open [328]), while others rely on a single dataset. This underscores the role of dataset diversity in improving generalization and robustness.

Physical Plausibility. In the absence of explicit physics constraints, generated videos may exhibit violations of physical realism, such as vehicle–background interpenetration, incorrect shadows, or scale distortions. While such issues may appear subtle in static frames, they significantly reduce realism when viewed as continuous video, undermining temporal coherence and physical plausibility.



Figure 8 Qualitative comparisons of state-of-the-art **VideoGen** models on the nuScenes [32] dataset. From top to bottom rows: Reference (from the dataset), MagicDrive [97], DreamForge [253], DriveDreamer-2 [464], and OpenDWM [312].

Controllability. Appearance-level controls (weather, time-of-day, style) can be reliably controlled via large-scale pre-trained video generation models with text conditioning. By contrast, precise geometric control over object position, orientation, and velocity remains challenging, typically requiring dedicated control embeddings or structured conditioning mechanisms.

Long-Tail Categories. Rare and small-scale classes (e.g., pedestrians, cyclists, traffic signs) remain hard to generate convincingly. Long-tail data imbalance often leads to unrealistic shapes, distorted geometry, or even omission of these critical objects.

Takeaways. The results suggest that future progress in video-based world models requires advances along five critical axes: (i) **realism**, reducing artifacts and enhancing detail fidelity; (ii) **consistency**, maintaining semantic and temporal coherence; (iii) **controllability**, unifying high-level appearance control with fine-grained geometric control; (iv) **physical plausibility**, incorporating physics priors to prevent unrealistic artifacts; and (v) **generalization**, leveraging diverse large-scale datasets to improve robustness.

4.4.2 Qualitative Analyses of OccGen Models

3D Geometric Consistency. Figure 9 shows qualitative results of occupancy generation models conditioned on scene layouts. The generated videos and occupancies exhibit strong spatial alignment across perspectives. Such cross-view coherence is crucial for maintaining geometric plausibility in multi-camera settings.

Occupancy Fidelity. The generated occupancies preserve key semantics, including drivable areas, sidewalks, and surrounding objects. While overall layouts are captured reliably for downstream perception, fine-grained geometry (e.g., thin lane boundaries, small dynamic agents) remains challenging, often leading to misalignment or incomplete reconstruction.

Controllability and Generalization. Conditioned on high-level scene priors, models can flexibly adapt to diverse intersection layouts and road structures, demonstrating promising controllability. However, rare structures and long-tail categories (e.g., bicycles, pedestrians) are often poorly represented, revealing limitations in data

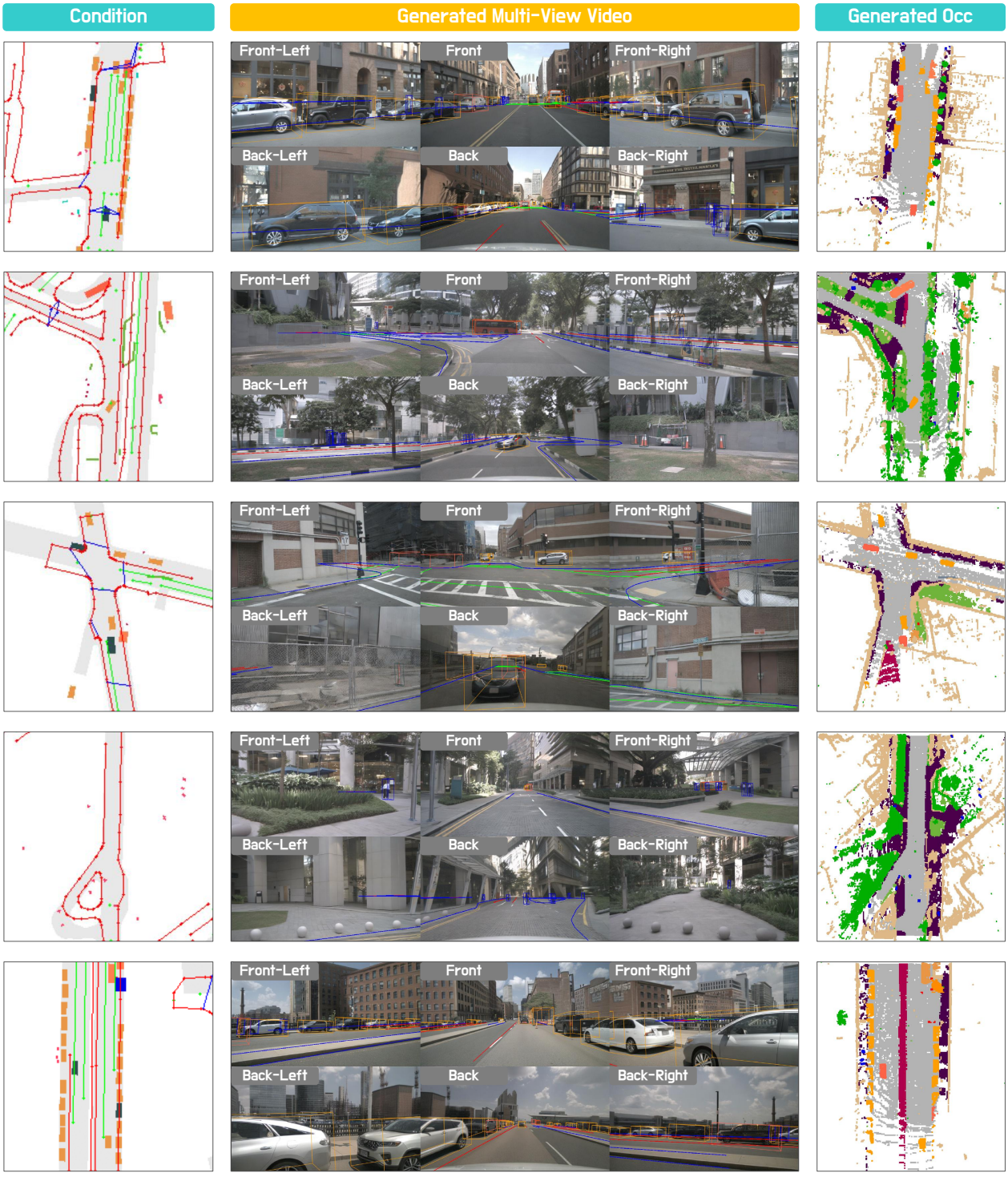


Figure 9 Qualitative examples of **OccGen** models on nuScenes [32]. From left to right columns: The input condition, the generated multi-view videos, and the generated occupancy grids. The results are generated using $\mathcal{X}$-Scene [425].

diversity and generalization capacity.

Takeaways. These results suggest that progress in occupancy generation hinges on three aspects: (i) **geometric consistency**, ensuring spatial coherence across 3D environments; (ii) **fine-grained fidelity**, particularly for small-scale and dynamic objects; and (iii) **generalization**, leveraging diverse datasets to handle rare layouts

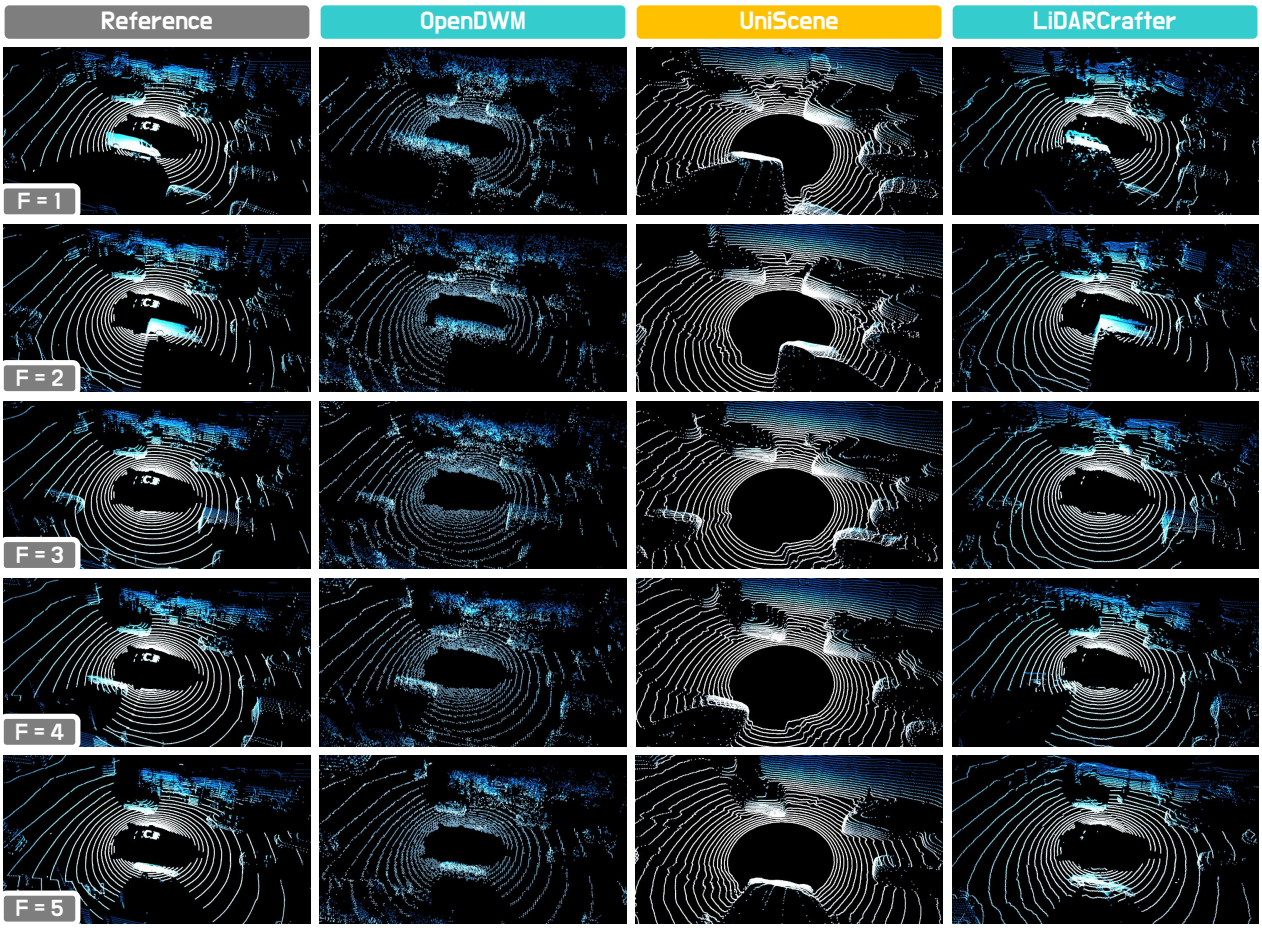


Figure 10 Qualitative comparisons of state-of-the-art **LiDARGen** models on the nuScenes [32] dataset. From left to right columns: Reference (from the dataset), OpenDWM [312], UniScene [181], and LiDARCrafter [205].

and long-tail classes. Advancing these aspects is essential for robust world models capable of supporting downstream tasks and closed-loop simulation.

4.4.3 Qualitative Analyses of LiDARGen Models

Global Patterns. Figure 10 compares representative LiDAR generation paradigms. The original scans exhibit dense rings with uniform angular spacing, faithfully capturing both static structures and dynamic objects. The voxel-based OpenDWM [312] emphasizes coherent scene geometry but often yields overly regularized patterns due to voxel-level modeling. The range-based LiDARCrafter [205] better preserves the native scanline structure with sharper rings, though it may introduce artifacts around occlusion boundaries. The occupancy-based UniScene [181] reproduces global distributions but tends to oversmooth fine details, leading to discontinuities.

Point Cloud Sparsity. Given the inherent sparsity of LiDAR data, generation models must balance realistic density with structural consistency. OpenDWM [312] often produces overly sparse regions, especially at long ranges. LiDARCrafter [205] maintains more uniform angular density, closely following the sensor's scanning characteristics. UniScene [181] provides globally complete coverage but sometimes introduces artificial filling inconsistent with real sensor patterns.

Object Completeness. Dynamic agents such as vehicles are particularly important for downstream perception and planning. OpenDWM [312] frequently underrepresents object contours, resulting in fragmented or partial shapes. LiDARCrafter [205] offers better surface completion, though finer details can be noisy. UniScene [181] reconstructs volumetrically plausible objects with consistent occupancy, but often lacks the sharp boundaries and crisp detail of real scans.

Takeaways. These results highlight three key attributes for LiDAR generation: (i) **global patterns**, ensuring coherent scene geometry while preserving sensor-specific scan structures; (ii) **point sparsity**, maintaining realistic density distributions that match LiDAR characteristics; and (iii) **object completeness**, accurately capturing dynamic agents with sharp contours and consistent surfaces. Future advances will require balancing these attributes to generate LiDAR sequences that are both perceptually realistic and physically faithful to sensor properties.

5 Applications

The versatility of 3D and 4D world models enables deployment across diverse domains. [1]**Autonomous Driving** (Sec. 5.1) supports simulation, evaluation, and scenario synthesis. [2]**Robotics** (Sec. 5.2) leverages them for navigation, manipulation, and scalable simulation. [3]**Video Games & XR** (Sec. 5.3) benefit from content generation, immersive rendering, and adaptive environments. [4]**Digital Twins** (Sec. 5.4) enable city-scale reconstruction, event replay, and scene editing. [5]**Emerging Applications** (Sec. 5.5) span scientific discovery, healthcare, industry, and disaster response. Together, these applications showcase the role of world models in unifying perception, prediction, and generation across domains.

5.1 Autonomous Driving

3D and 4D world models provide a principled foundation for autonomous driving, supporting simulation, evaluation, and scenario synthesis. They enable controllable, interactive, and safety-critical environments that cannot be easily reproduced in the real world. We outline three major applications.

Traffic Simulation. World models enable realistic traffic simulators with heterogeneous agents, diverse motion, and physics-compliant interactions. Compared with image-only platforms, volumetric representations such as occupancy grids [355, 444], multi-frame LiDAR point clouds [205], or scene-level meshes [258] provide richer geometry and temporal coherence [368, 429]. Modern systems further support controllable parameters (*e.g.*, traffic density, intent, weather) and stochastic perturbations, improving robustness and generalization for downstream policies [127, 307, 351, 365, 375, 376, 458, 464].

Closed-Loop Driving Evaluation. Beyond static benchmarks, closed-loop setups couple generative models with agents to assess *perception*→*planning*→*control* stacks over long horizons [429, 463]. By jointly modeling ego behavior and surrounding traffic dynamics, models create responsive environments that adapt to agent actions in real time [43, 205]. This allows scalable evaluation of robustness under distribution shifts, rare events, and recovery after failures [307, 458], while modular conditioning (*e.g.*, HD maps, text queries, and ego trajectories) enables targeted stress testing [43, 96]. Recent frameworks tighten this loop: AD-R1 [410] couples end-to-end driving with closed-loop reinforcement learning over an impartial world model, while Xiaomi OneVL [240] performs one-step latent reasoning and planning with vision-language explanations. VITAL [204] connects visual motion to driving actions through a coarse metric trajectory scaffold and a compact residual encoding its remaining error. Future observations supervise this representation during training, while a joint action expert predicts both components in one step from current observations, ego state, and navigation instructions at inference, providing a direct world-to-action interface without generating future images.

Scenario Synthesis. World models can generate rare or safety-critical driving scenes that are underrepresented in real datasets, which is essential for evaluating robustness. Typical cases include severe occlusions, sudden intrusions, multi-agent conflicts, and adverse weather [140, 229, 490]. Controllable generation with HD maps, semantic masks, scene graphs, or textual prompts enables targeted testing [43, 205, 490]. Physics- and motion-aware models ensure dynamic feasibility [126, 479], while stochastic sampling improves coverage of rare events. LiDAR-centric approaches such as LiDARCrafter [205] further extend this capability to 4D sequences with temporal coherence.

5.2 Robotics

3D and 4D world models have the potential to enhance robotic intelligence by supporting navigation, manipulation, and simulation. They provide spatial-temporal grounding, physical reasoning, and scalable synthetic environments, which are crucial for robust policy learning.

Embodied Navigation. Robots leverage world models to perceive and predict dynamic layouts, enabling long-horizon exploration, obstacle avoidance, and localization in both structured and unstructured settings [158, 332, 481]. Forecasting future states is critical in crowded or occluded scenes [158, 389], where multi-scan LiDAR, voxelized occupancy, and predictive dynamics provide reliable spatial-temporal cues [89, 429]. Recent studies also combine visual, topological, and linguistic signals for instruction following and adaptive decision-making [137, 481]. A growing line of work casts navigation itself as world modeling: NavThinker [132] couples action-conditioned prediction with planning for social navigation, language-conditioned world models support visual navigation [77], memory-augmented planning and foresight move toward unified navigation world models [76], and FLUX [103] accelerates cross-embodiment generative navigation policies via rectified flow.

Object-Centric Manipulation. For this task, models capture object geometry and physical transitions, allowing robots to anticipate contact dynamics and plan stable grasps or rearrangements [20, 248, 484]. Representations such as meshes, keypoint graphs, and volumetric embeddings support fine-grained control and generalization to new objects [161, 264]. Integration of differentiable physics with generative models yields physically consistent predictions that can be optimized for various tasks [20, 82, 484]. Recent foundation models target manipulation directly, including PAIWorld [142], a 3D-consistent world foundation model for robotic manipulation, and Qwen-RobotWorld [446], a language-conditioned video world model for embodied robot control.

DepthWorld [18] strengthens geometric supervision for manipulation through DROID-3D, which combines learned stereo depth with robot-grounded multi-view calibration. Its action-conditioned video model jointly predicts RGB and depth through spatial latent tiling and robot-frame point-map supervision. Held-out-trajectory results support improved visual and geometric prediction, while the same-robot, same-lab split leaves cross-embodiment generalization untested.

Scene Generation for Simulation. Generative models create diverse synthetic environments, reducing manual design costs for training and evaluation [4, 158, 315]. Procedural variation in layout, semantics, and dynamics exposes robots to a wide range of scenarios, improving robustness and sim-to-real transfer [82, 332, 481, 484]. Flexible scene representations, from meshes to voxel grids and point clouds, further enable integration with both physics-based simulators and photorealistic renderers [20, 264].

5.3 Video Games & XR

World models transform gaming and XR by automating content creation, supporting immersive rendering, and enabling adaptive environments that respond to player actions.

Procedural World Generation. Generative models automate the design of expansive virtual worlds, supporting open exploration and emergent gameplay [66, 153, 257]. Procedural pipelines can incorporate maps, player states, or language prompts to scale content production beyond manual asset creation [129, 257]. Maintaining temporal and semantic coherence is key for believable dynamic evolution [101], while diverse scene representations such as point clouds, voxels, and neural radiance fields balance realism, style, and efficiency [165, 345].

Interactive Scene Rendering. Immersive XR requires real-time rendering of dynamic scenes where users move freely through evolving geometry and lighting [189, 427]. Neural representations including NeRF [345] and Gaussian Splatting [165] advance photorealistic synthesis, with temporal extensions modeling motion and state change [9, 94]. To ensure consistency and comfort, systems must maintain geometric fidelity under arbitrary viewpoints, adapt scene content to user actions [13, 438], and employ efficient pipelines to sustain high frame rates.

Playable Environment Adaptation. Adaptive worlds adjust geometry, layout, and agent behavior to sustain challenge and engagement [66, 99, 435, 437]. 3D/4D models support real-time transformations such as altering terrain, collapsing structures, or spawning entities based on player interactions [129, 437]. By leveraging priors or high-level instructions, these systems preserve style, physics, and narrative coherence [101, 205], thereby enhancing immersion, replayability, and personalized gameplay. Emerging interactive world models push these capabilities further, including Matrix-Game 3.0 [371], a real-time streaming model with long-horizon memory; Hunyuan-GameCraft-2 [334], an instruction-following game world model; Solaris [311], a multiplayer video world model demonstrated in Minecraft; and 3D4D [121], an interactive and editable 4D world model built on 3D video generation. Extending this line from generation toward control, GameWAM [115] reports a

world-action model for native closed-loop gameplay and GUI control, arguing that games uniquely combine first-person perception, rapid visual change, persistent world state, and heterogeneous native control schemes — a combination under which prior game agents map observations straight to actions without modelling dynamics, while interactive game world models predict visuals without acting.

5.4 Digital Twins

3D and 4D world models underpin urban digital twins by enabling large-scale reconstruction, event replay, and interactive editing. These capabilities support planning, analysis, and simulation in smart city applications.

City-Scale Scene Modeling. Digital twins integrate multimodal sensing, including LiDAR, RGB-D, aerial photogrammetry, and drone surveys, to capture both static infrastructure and dynamic activities [24, 210, 394]. They enable applications such as traffic monitoring, infrastructure planning, and disaster response [21, 314], while dynamic modeling simulates pedestrian and vehicle flows for capacity planning [333, 335]. Recent advances in streaming pipelines and 4D compression maintain temporal consistency and allow metropolitan-scale deployment [67, 72]. Complementary efforts pursue language-driven generation of urban environments, such as MajutsuCity [143], which enables controllable city-scale 3D generation with editable assets and layouts.

Event Replay & Forecasting. World models reconstruct past or hypothetical events from sparse sensor logs, aiding analysis of incidents [63, 128], construction monitoring [349], or emergency response [110]. Replayable 4D scenes clarify causality, while predictive extensions enable what-if simulations for evaluating interventions. Alignment with sensor ground truth remains critical for reliability.

Scene Control & Editing. Interactive tools allow users to manipulate urban content for simulation and visualization, including vehicle removal, weather alteration, and layout modification [67, 101]. Such controllability improves planning workflows and supports immersive city-scale analysis.

5.5 Other Emerging Applications

Beyond autonomous driving and robotics, 3D and 4D world models are expanding into scientific, medical, industrial, and safety-critical domains. These applications highlight their versatility in modeling complex spatial–temporal systems. General-purpose world foundation models increasingly aim to serve these domains at once, including Cosmos 3 [280], a family of omnimodal world foundation models for physical AI; PAN [337], a general-purpose, interactive, long-horizon world model; HY-World 2.0 [148], a multimodal model for reconstructing, generating, and simulating 3D worlds; and Lyra 2.0 [316], which produces explorable generative 3D worlds.

Scientific Discovery and Environmental Modeling. World models capture natural dynamics from multimodal observations, supporting forecasting and exploratory simulation. Applications include climate and weather prediction [23, 177, 285], monitoring glacier retreat or floods, and simulating wildfire spread. By learning directly from data, they complement physics-based solvers with faster iteration.

Healthcare & Biomechanics. Generative 3D models reproduce anatomy deformation and tissue behavior for surgical training, planning, and guidance [424]. Predictive motion models aid rehabilitation, prosthetics, and injury prevention by anticipating joint trajectories [168], enhanced by multi-view capture and volumetric reconstruction. Dedicated medical world models [218] extend this direction by modeling clinical state and dynamics for intervention planning.

Industrial Process & Manufacturing Simulation. Virtual prototyping with world models supports robotic assembly, material handling, and inspection [28, 58]. Temporal simulation of component interactions reduces costly trials and enables analysis of efficiency and fault recovery.

Security, Defense & Disaster Response. Synthetic environments simulate tactical operations, hazardous conditions, and evacuations [347]. Dynamic scene modeling further aids disaster preparedness by predicting structural collapse, fire spread, or chemical dispersion, and testing emergency response plans.

6 Challenges & Future Directions

In this section, we highlight key challenges of world models, including benchmarking, long-horizon fidelity, the transition from pseudo-4D to continuous 4D dynamics, physical realism, efficiency, and cross-modal coherence, and outline directions for future research.

6.1 Standardized Benchmarking & Evaluations

A major barrier to progress in driving world models is the absence of common, standardized benchmarks and protocols. Because studies rely on different datasets and ad hoc metrics, meaningfully comparing models across diverse, realistic settings is difficult [80, 144, 185, 468]. A unified benchmark would instead jointly capture the qualities that matter most, such as physical plausibility, temporal consistency, and controllability.

Recent efforts begin to move in this direction. 4DWorldBench [243] consolidates fidelity, consistency, and controllability into a single framework spanning 3D and 4D generation models, while WorldLens [203] operationalizes holistic, multi-aspect evaluation and shows that no current approach dominates across pixel quality, 4D geometry, closed-loop behavior, and human realism, underscoring the gap between how generated worlds *look* and whether they *behave*. Building on such protocols, future evaluation should move beyond perceptual scores such as FID/FVD toward explicit *physics- and causal-consistency* criteria (*e.g.*, collision and kinematic plausibility, and action-conditioned counterfactual probes; *cf.* Sec. 4.2.6), ideally backed by a shared open-source codebase for protocol-level comparability.

Equally important is grounding evaluation in behavior rather than appearance alone. Standardized protocols should span both closed-loop simulation and real-world scenarios, stressing models under varying traffic densities, weather conditions, and complex urban layouts [423]. A growing family of closed-loop benchmarks moves toward this goal: World-in-World [445] evaluates world models within an interactive closed loop, DriveE2E [436] and nuPlan-R [288] provide real-to-simulation and reactive multi-agent planning benchmarks, ReactSim-Bench [459] probes reactive behavior, and DrivingGen [487] consolidates generative-video-model evaluation. A complementary concern is trusting the simulator itself: its verdict should be *validated* before it informs decision-making [281]. Developing these benchmarks is essential for fair, transparent comparison across approaches.

6.2 High-Fidelity & Long-Horizon Generation

Another critical challenge is achieving high-fidelity generation over long time horizons [48, 253]. Short-term predictions capture immediate interactions reasonably well, but small errors accumulate over longer sequences, gradually producing unrealistic behavior and eroding scene consistency.

A primary root cause is the mismatch between *open-loop teacher forcing* at training and *closed-loop autoregressive* inference: once a model conditions on its own previously generated outputs, the input distribution drifts from the training distribution, inducing covariate shift and compounding errors over the rollout. Current work mitigates this through scheduled sampling and rollout or noise augmentation that expose the model to its own predictions, diffusion-forcing and chain-of-forward training [449], explicit memory or history re-anchoring [114], and self-consistency objectives. Yet robust long-horizon stability remains open, compounded by dynamic urban environments in which multiple agents and environmental factors evolve continuously.

Addressing this will require training paradigms [449] and memory mechanisms [114] that explicitly penalize long-term divergence for reliable extended simulation. Encouragingly, progress spans modalities: action-conditioned generators such as PerpetualWonder [443] sustain coherent 4D scene generation over far longer horizons, AutoScape [46] enforces geometry consistency to extend scene generation, and OccSim [224] pushes occupancy world models to multi-kilometer rollouts, showing long-horizon stability being attacked from both the video and geometric-representation sides.

6.3 From Pseudo-4D to Continuous 4D Dynamics

Much existing work realizes the temporal axis by stacking discrete 3D frames, yielding what is effectively *pseudo-4D*: adjacent snapshots stitched together rather than a continuous spatiotemporal field. Such stacking

does not guarantee dynamical continuity, often manifesting as flicker, geometric drift, or physically implausible motion between frames. Recent analysis attributes part of this to a *seriality gap* in video diffusion models, where frames are produced in parallel without modeling sequential dynamics [42].

A cluster of recent reconstruction work supplies concrete machinery for this transition. DAV4 [84] applies Riemannian flow matching directly to 4D Gaussian Splatting parameters, defining probability paths on the non-Euclidean manifolds of scale, rotation, and opacity so that every intermediate state remains a valid Gaussian — a constraint that frame-stacking formulations never have to satisfy. UniQuery4R [52] makes time a decoding-time query rather than an index over precomputed frames, encoding a clip once and selecting source view, target view, and continuous coordinate only when decoding. Gallileo-4D [310] pursues the same target with a frozen-backbone ensemble, and SceneReGen [340] attacks the complementary problem of placing generatively completed objects into a shared observation-aligned frame instead of the object-centred, scale-normalized frame that object-level generative priors natively emit. Together these indicate that continuous 4D is becoming a property of the representation and its query interface rather than something recovered post hoc from stacked snapshots.

Reaching *true continuous 4D* will likely require continuous-time formulations (*e.g.*, neural-ODE-style or flow-based dynamics), deformation- and scene-flow-aware representations, and explicit physical or kinematic constraints, so that worlds evolve along coherent, physically consistent trajectories rather than sampled keyframes. Early steps point this way: ray-centric sequence models treat 4D LiDAR as a coherent temporal stream [222], feed-forward trajectory-field formulations reconstruct any-view, any-time 4D geometry [157], and occupancy-guided generation anchors video synthesis to an evolving 4D field [431]. Stream4D [16] tackles the streaming setting directly, arguing that training objectives optimized for local next-frame prediction are precisely what let geometric drift accumulate over long autoregressive rollouts, and proposing 4D-consistency as an explicit training signal rather than an emergent property of frame-level accuracy.

6.4 Physical Fidelity, Controllability & Generalizability

From the perspective of generation capability, current driving world models are critically limited along three axes: *physical realism*, fine-grained *controllability*, and robust *generalization* [96, 432]. On physical realism, they often produce implausible events, such as non-deforming collisions and temporally inconsistent objects [163]. Recent work targets this explicitly, *e.g.*, physics-aware 4D scene generation that bakes physical interactions into the process [491], and causality-aware formulations that model the cause–effect structure of driving dynamics rather than surface correlations [59, 412].

Controllability is similarly coarse: editing is typically confined to traffic agents' positions or appearances, with limited control over environmental elements such as architecture or road signs. Emerging systems widen this surface, including fully editable driving worlds [75], editable semantic-occupancy scenes [14], and point-conditioned object insertion [156].

Physical realism is also being pursued by constraining the dynamics rather than by scaling the generator. LagrangeGS [309] formulates dynamic 3D Gaussian Splatting as a non-conservative Lagrangian system, on the objection that existing physics-aware extensions merely fit velocity fields to observed visual deformation without satisfying any Lagrangian structure, which is precisely why their extrapolation degrades outside the observed interval. VersaGauss [324] broadens the range of dynamics a Gaussian representation can express, targeting multiphase phenomena. Both suggest that the route to physical compliance runs through the equations of motion the representation is required to obey, and that such compliance should be measured by extrapolation beyond the fitted window rather than by reconstruction error inside it.

Most critically, these models tend to overfit training data and fail to generalize to new environments and rare objects, limiting real-world applicability; generative scaling of long-tail scenarios [220] offers one route to broadening coverage. Overcoming these three limitations jointly is a prerequisite for faithful, controllable, and generalizable world models.

6.5 Computational Efficiency & Real-Time Performance

A further limitation lies in computational efficiency and real-time responsiveness. Existing methods often depend on heavy architectures and multi-step sampling, incurring latency and memory overhead that undermine

large-scale generation and simulation. Progress calls for sparse computation [304] and inference-acceleration techniques [45] that keep models accurate and responsive while remaining scalable.

This goal is increasingly within reach. Early real-time systems such as INSPATIO-WORLD [149] perform spatiotemporal autoregressive 4D simulation at interactive rates, reinforced by real-time closed-loop simulators [279] and streaming interactive world models with long-horizon memory [371]. Efficiency-oriented designs such as sparse, dynamically queried 4D occupancy models [64] further reduce compute without sacrificing coverage.

6.6 Cross-Modal Generation Coherence

Current world models often struggle with consistent cross-modal generation, in which visual, geometric, and semantic modalities must jointly form a coherent representation of the environment. Misalignment can produce imagery that conflicts with the underlying 3D structure, undermining downstream perception and planning. A related failure mode appears when converting 3D engine renderings, such as depth maps and untextured geometry, into photorealistic video for gaming and immersive content: long-horizon autoregressive generation can drift away from the geometry it was conditioned on. Closing the Loop [247] addresses this training-free, enforcing revisit consistency so that returning to a previously rendered viewpoint yields a coherent, not merely plausible, result.

Overcoming this requires integrated architectures that learn jointly from diverse sensor data while enforcing strict consistency during generation [181, 396]. Recent unified models point the way: joint reconstruction-and-generation world models [478] and Gaussian-centric driving world models coupling scene understanding with multi-modal generation [69] share a single latent representation across modalities, keeping visual, geometric, and semantic outputs consistent. Fine-grained spatial alignment and temporal synchronization across streams remain crucial for dynamic driving interactions, and future research should target this fundamental challenge.

7 Conclusion

This survey has presented the first systematic review of **3D and 4D world modeling and generation**, clarifying definitions, organizing methods into a hierarchical taxonomy across VideoGen, OccGen, and LiDARGen, and summarizing datasets, evaluations, and applications. By shifting focus from purely visual realism to geometry-grounded modeling, native 3D and 4D representations enable models to achieve plausibility, controllability, and physical consistency, serving roles as data engines, action interpreters, neural simulators, and scene reconstructors. Despite rapid progress, challenges remain in scaling to real-world complexity, aligning multimodal signals, and establishing standardized evaluation for controllability, safety, and generalization. Looking forward, unifying generative and predictive paradigms, integrating language and reasoning, and advancing simulation and digital twin ecosystems represent promising directions. Equally important will be community efforts in creating open benchmarks, reproducible codebases, and large-scale datasets tailored for 3D/4D world models, which can accelerate progress and ensure comparability across methods. We hope this survey provides both a coherent foundation and a forward-looking roadmap for advancing robust, interpretable, and generalizable world models to power the next generation of embodied AI.

Table 15 Summary of the evaluation metrics used for evaluating the quality of [1]**Generation**, [2]**Forecasting**, [3]**Planning**, [4]**Reconstruction**, and [5]**Downstream Tasks** for the **VideoGen**, **OccGen**, and **LiDARGen** models in 2D, 3D, and 4D tasks.

Abbr.	-	Full Name	Description	Ref.
⚖ **Generation Quality - Perceptual Fidelity**				
FID	↓	Fréchet Inception Distance	Statistical distance between multivariate Gaussians fitted to Inception features of real and generated samples, measuring distributional similarity.	[122]
FVD	↓	Fréchet Video Distance	Statistical distance between multivariate Gaussians fitted to I3D features [40] of real and generated videos, capturing temporal coherence.	[344]
FRD	↓	Fréchet Range Distance	Statistical distance between Gaussians fitted to RangeNet++ features [260] extracted from LiDAR range images, assessing distributional fidelity.	[269]
FPD	↓	Fréchet Point Cloud Distance	Statistical distance between Gaussians fitted to PointNet features [295] of raw 3D point clouds, evaluating geometric realism.	[320]
FSVD	↓	Fréchet Sparse Volume Distance	Statistical distance between Gaussians fitted to volumetric encoder features of sparse voxel inputs, capturing volumetric structure.	[300]
FPVD	↓	Fréchet Point Volume Distance	Statistical distance between Gaussians fitted to volumetric encoder features of hybrid point–voxel representations, measuring fidelity.	[300]
F3D	↓	Fréchet 3D Distance	Statistical distance between Gaussians fitted to occupancy grid features, evaluating volumetric realism in generated 3D data.	[230]
S-FRD	↓	Semantic Fréchet Range Distance	Class-aware extension of FRD that incorporates semantic labels for improved alignment of LiDAR range distributions.	[490]
S-FPD	↓	Semantic Fréchet Point Distance	Class-aware extension of FPD that integrates semantic labels to assess alignment of 3D point cloud distributions.	[490]
KID	↓	Kernel Inception Distance	Maximum Mean Discrepancy between Inception features using a polynomial kernel, providing an unbiased distributional similarity measure.	[26]
IS	↑	Inception Score	Evaluates image realism by rewarding confident and diverse class predictions from a pretrained Inception classifier, without real reference data.	[308]
IQ	↑	Image Quality	Predicts perceptual image quality by estimating human opinion scores with a learned quality assessor [164], without ground truth references.	[154]
⚖ **Generation Quality - Statistical Fidelity**				
PR	↑	Precision-Recall	Reports sample fidelity as precision and distributional coverage as recall, characterizing closeness to the real data manifold.	[24]
SWD	↓	Sliced Wasserstein Distance	Mean Wasserstein distance over multiple random 1D projections of image patches at different scales, reflecting distributional similarity.	[116]
JSD	↓	Jensen–Shannon Divergence	Symmetric divergence measuring similarity between occupancy histograms of real and generated scenes, lower indicating better alignment.	[499]
MMD	↓	Minimum Matching Distance	Average Chamfer distance from each real sample to its nearest generated neighbor, quantifying geometric fidelity.	[499]
COV	↑	Coverage	Fraction of real samples matched by at least one generated output, measuring generative diversity and recall.	[172]
1-NNA	-	1-Nearest-Neighbor Accuracy	Overlap test using a 1-NN classifier trained across sets, where accuracy near 50% indicates distributional equivalence.	[172]
Diversity	↑	-	Degree of variability across generated outputs for fixed prompts, often measured via pixel- or feature-wise variance.	[488]
⚖ **Generation Quality - Spatial Consistency**				
VCS	↑	View Consistency Score	Summation of LoFTR keypoint confidences [326] across overlapping views, evaluating multi-view geometric consistency and alignment quality.	[330]

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Abbr.	-	Full Name	Description	Ref.
KPM	↑	Key Points Matching	Ratio of successfully matched keypoints between adjacent generated and real views, reflecting geometric alignment quality.	[368]
DAS	↓	Depth Alignment Score	Statistical discrepancy between projected point clouds and estimated monocular depth [421], measuring scene-level depth consistency.	[396]
⚖ **Generation Quality - Temporal Consistency**				
CTC	↑	CLIP Temporal Consistency	Cosine similarity of CLIP features [299] across consecutive frames, measuring temporal stability and smoothness in generated video sequences.	[154]
DTC	↑	DINO Temporal Consistency	Cosine similarity of DINO features [282] across adjacent frames, evaluating temporal coherence and smooth transitions in generated sequences.	[154]
TTCE	↓	Temporal Transformation Consistency Error	Registration error between temporally generated and ground-truth point clouds, evaluating motion alignment and temporal consistency.	[205]
CTC	↓	Chamfer Temporal Consistency	Chamfer distance between generated point clouds across different timestamps, quantifying temporal stability and geometric coherence.	[205]
ICP	↓	ICP Energy / Outlier	Registration residuals and outlier ratios from Iterative Closest Point alignment of LiDAR frames, detecting temporal jitter and misalignment.	[500]
⚖ **Generation Quality - Subject Consistency**				
SC	↑	Subject Consistency	Cosine similarity of subject-region features [282] across frames, evaluating identity persistence and stability in generated video sequences.	[144]
FDC	↑	Foreground Detection Confidence	Confidence scores of detected foreground objects in generated samples using a pretrained detector, reflecting semantic plausibility and realism.	[205]
CFCA	↑	Conditional Foreground Classification Accuracy	Semantic consistency of generated objects evaluated by classification accuracy with a pretrained object classifier, conditioned on ground truth.	[205]
CFSC	↑	Conditional Foreground Spatial Consistency	Mean IoU between 3D boxes regressed by a conditional VAE and ground truth boxes, assessing geometric alignment under conditioning.	[205]
⚖ **Generation Quality - Controllability**				
CDA	↑	Conditional Detection Accuracy	Standard detection accuracy from a pretrained 3D detector applied to generated point clouds with box conditioning, measuring semantic fidelity.	[205]
CLIP-Sim	↑	CLIP Similarity	Average cosine similarity between CLIP embeddings of generated and reference frames, reflecting semantic alignment across modalities.	[245]
MAE	↓	Mean Absolute Error	Difference in predicted versus reference object counts within scene graphs, assessing accuracy of graph-level controllability.	[231]
JI	↑	Jaccard Index	Overlap ratio of predicted and reference category sets within scene graphs, evaluating graph-level semantic consistency.	[231]
RotErr	↓	Rotation Error	Angular difference between recovered and target camera trajectories, quantifying rotational alignment error.	[160]
TransErr	↓	Translation Error	Euclidean distance between recovered and target camera trajectories, quantifying translational alignment error.	[160]
⚖ **Generation Quality - Human Preference**				
VQ	↑	Visual Quality (2AFC)	Win rates for perceptual visual quality in two-alternative forced choice comparisons, reflecting human-preferred realism of generations.	[363]

Note: Continued on next page

Abbr.	-	Full Name	Description	Ref.
MR	↑	Motion Rationality (2AFC)	Win rates for perceived motion rationality in two-alternative forced choice settings, evaluating naturalness of temporal dynamics.	[363]
DMOS	↑	Differential Mean Opinion Score	Average human-rated alignment with conditioning constraints (e.g., scene graphs), providing a relative perceptual quality measure.	[231]
⚖ **Forecasting Quality - Spatial Predictive Accuracy**				
L1 Error	↓	Frame Mean Absolute Error	Pixel- or depth-space L1 distance between predicted and ground-truth frames, quantifying reconstruction fidelity and short-horizon accuracy.	[386]
L2 Error	↓	Frame Mean Squared Error	Pixel- or depth-space L2 distance between predicted and ground-truth frames, reflecting average squared reconstruction deviation.	[386]
$\mathbf{IoU}_{\mathrm{c}}$	↑	IoU at Current Timestamp	Intersection-over-Union between predicted and reference occupancy maps at the current frame, assessing immediate prediction quality.	[246]
$\mathbf{IoU}_{\mathrm{f}}$	↑	IoU at Future Timestamp	Intersection-over-Union between predicted and reference occupancy maps at a fixed future horizon, capturing long-range prediction quality.	[246]
$\mathbf{IoU}_{\mathrm{wf}}$	↑	IoU at Weighted Future Timestamp	Weighted average Intersection-over-Union across multiple future frames, emphasizing near-term predictions for smoother accuracy assessment.	[246]
CD	↓	Chamfer Distance	Bidirectional nearest-neighbor distance between point clouds from ray-cast predicted and ground-truth occupancy, measuring geometric fidelity.	[451]
L1 Med	↓	Median L1 Depth Error	Median absolute depth error along LiDAR rays after projection, robustly quantifying accuracy against outliers.	[451]
AbsRel Med	↓	Median Absolute Relative Error	Median of relative depth errors across all LiDAR rays, providing a scale-aware and robust measure of accuracy.	[451]
L1 Mean	↓	Mean L1 Depth Error	Mean absolute depth error along projected LiDAR rays, reflecting average deviation in meters from reference.	[451]
AbsRel Mean	↓	Mean Absolute Relative Error	Mean of relative depth errors across all rays, capturing overall scale-consistent accuracy of depth predictions.	[451]
⚖ **Forecasting Quality - Temporal Predictive Accuracy**				
KODP	↑	Key Object Dimension Probability	Probability-based measure penalizing implausible object dimensions using category priors, encouraging physically realistic and consistent generation.	[367]
TFSC	↑	Temporal Foreground Shape Consistency	Voxel-level Intersection-over-Union of dynamic object instances across consecutive frames, ensuring shape persistence and temporal stability.	[367]
TBEC	↑	Temporal Background Environment Consistency	Consistency of static voxels under ego-motion compensation, validating environmental rigidity and long-term background stability.	[367]
⚖ **Planning Quality - Open-Loop Planning**				
ADE	↓	Average Displacement Error	Mean displacement error between predicted trajectories and expert waypoints across the horizon, reflecting overall trajectory accuracy.	[418]
FDE	↓	Final Displacement Error	Displacement error at the final predicted waypoint compared with expert trajectories, emphasizing long-term accuracy.	[418]
SLE	↓	Speed L1 Error	Mean absolute error of predicted speed control signals.	[152]
SALE	↓	Steer Angle L1 Error	Mean absolute error of predicted steering angle control signals.	[152]
CR	↓	Collision rate	Fraction of rollouts in which the controlled vehicle collides with surrounding agents or obstacles, indicating safety risk.	[368]

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Abbr.	-	Full Name	Description	Ref.
PDMS	↑	Predictive Driver Model Score	Aggregate score combining progress, spacing, and comfort after discarding unsafe rollouts, approximating human-like driving quality.	[65]
EPDMS	↑	Extended Predictive Driver Model Score	Extended version of PDMS that includes nine additional factors to reflect rule adherence and recovery behaviors.	[39]
AHE	↓	Average Heading Error	Mean absolute angular deviation between predicted and expert heading over the trajectory horizon, measuring orientation accuracy.	[33]
FHE	↓	Final Heading Error	Absolute angular deviation of predicted heading from expert at the final timestep, reflecting terminal orientation accuracy.	[33]
MR	↓	Miss Rate	Fraction of prediction timesteps where displacement error exceeds horizon-specific thresholds, reflecting failure in trajectory coverage.	[33]
⚖ **Planning Quality - Closed-Loop Planning**				
SR	↑	Success Rate	Percentage of navigation episodes that successfully reach the goal within a fixed time budget, indicating overall task completion.	[78]
ID	↑	Infraction Distance	Average driving distance between two infractions, with longer distances reflecting safer and more reliable policy behavior.	[78]
ADS	↑	Arena Driving Score	Composite score combining route completion metrics with PDMS to summarize closed-loop driving performance in Arena environments.	[423]
NAC	↑	No At-Fault Collisions	Fraction of scenarios without ego-fault collisions, focusing exclusively on responsibility-aware collision evaluation.	[33]
DAC	↑	Drivable-Area Compliance	Boolean evaluation that checks whether the ego vehicle remains inside drivable polygons throughout the rollout.	[33]
DDC	↑	Driving-Direction Compliance	Boolean evaluation verifying that ego motion remains aligned with the designated lane's legal driving direction.	[33]
MP	↑	Making Progress	Boolean check confirming that the ego vehicle makes sufficient forward route progress within the evaluation horizon.	[33]
TTC	↑	Time-to-Collision	Boolean verification that the time-to-collision value exceeds safety thresholds, preventing imminent crashes.	[33]
PAR	↑	Progress Along Route	Ratio of ego-vehicle progress compared to expert trajectory progress along the same route, reflecting efficiency.	[33]
SLC	↑	Speed-Limit Compliance	Score penalizing magnitude and duration of speed-limit violations, higher values indicating safer speed adherence.	[33]
Comfort	↑	Driving Comfort	Penalization of excessive jerk, acceleration, or yaw-rate, reflecting ride quality and passenger comfort.	[33]
⚖ **Reconstruction Quality - Photometric Fidelity**				
PSNR	↑	Peak Signal-to-Noise Ratio	Logarithmic ratio of maximum possible signal power to reconstruction error, measuring pixel-level fidelity of generated images	[146]
SSIM	↑	Structural Similarity Index Measure	Quality index considering structural similarity, luminance consistency, and contrast preservation between generated and reference images.	[370]
LPIPS	↓	Learned Perceptual Image Patch Similarity	Feature-space distance between deep network activations of image patches, quantifying perceptual realism beyond pixel fidelity.	[454]
⚖ **Reconstruction Quality - View Changing Consistency**				
NTA-IoU	↑	Novel Trajectory Agent IoU	Intersection-over-Union between projected 3D bounding boxes of foreground agents and detected 2D boxes under novel viewpoints.	[463]
NTL-IoU	↑	Novel Trajectory Lane IoU	Intersection-over-Union between projected lane structures and detected lane markings from novel viewpoints, evaluating background alignment.	[463]
⚖ **Downstream Evaluation - Detection**				

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Abbr.	-	Full Name	Description	Ref.
mAP	↑	Mean Average Precision	Average precision computed over multiple IoU thresholds for 2D detection boxes on standard benchmarks, reflecting detection accuracy.	[212]
mAP-3D	↑	Mean Average Precision in 3D	Average precision for 3D bounding boxes, integrating precision-recall across multiple IoU thresholds in 3D space.	[32]
LET-3D-AP	↑	Average Precision with Longitudinal Error Tolerance	3D average precision using LET-IoU, allowing depth shifts along the camera ray within a tolerance margin.	[145]
LET-3D-APL	↑	Longitudinal Affinity Weighted LET-3D-AP	Weighted LET-3D-AP that penalizes larger longitudinal corrections, improving realism in depth-sensitive evaluation.	[145]
mATE	↓	Mean Average Translation Error	Mean L2 distance between predicted and ground-truth object centers for matched true positives, evaluating localization accuracy.	[32]
mASE	↓	Mean Average Scale Error	Mean scale discrepancy defined as one minus IoU for matched true positives, reflecting object size accuracy.	[32]
mAOE	↓	Mean Average Orientation Error	Mean absolute yaw error of matched true positives, quantifying accuracy of predicted object orientation.	[32]
mAVE	↓	Mean Average Velocity Error	Mean L2 difference between predicted and ground-truth object velocities for matched true positives, measuring motion accuracy.	[32]
mAAE	↓	Mean Average Attribute Error	Mean error in predicting semantic attributes for matched true positives, computed as one minus attribute accuracy.	[32]
NDS	↑	nuScenes Detection Score	Composite metric combining mAP with normalized true positive errors (mATE, mASE, mAOE, mAVE, mAAE), reflecting holistic detection quality.	[32]
⚖ **Downstream Evaluation - Segmentation**				
mIoU	↑	Mean Intersection over Union	Average Intersection-over-Union across all semantic classes in 2D images or 3D point clouds, measuring segmentation accuracy.	[83]
BEV-Map-IoU	↑	Bird's-Eye-View Map IoU	Class-wise Intersection-over-Union for freespace, lane, and dynamic agents in BEV compared with HD maps, evaluating scene consistency.	[291]
⚖ **Downstream Evaluation - Tracking**				
MOTA	↑	Multi-Object Tracking Accuracy	Composite metric aggregating false positives, false negatives, and identity switches into a single score, evaluating overall tracking reliability.	[22]
MOTP	↑	Multi-Object Tracking Precision	Mean Intersection-over-Union between matched predictions and ground truth across frames, quantifying spatial localization precision.	[22]
3D-AMOTA	↑	Average Multi-Object Tracking Accuracy in 3D	3D extension of MOTA averaged across recall thresholds, reducing sensitivity to threshold choices while evaluating tracking accuracy.	[377]
3D-AMOTP	↑	Average Multi-Object Tracking Precision in 3D	3D extension of MOTP averaging localization precision (IoU or center distance) over recall thresholds, reflecting robustness.	[377]
⚖ **Downstream Evaluation - Occupancy Prediction**				
Occupancy-IoU	↑	Occupancy Intersection over Union	Intersection-over-Union between predicted and labeled voxel occupancies at class- or scene-level granularity, reflecting occupancy accuracy.	[323]
VPQ	↑	Voxelized Panoptic Quality	Panoptic quality metric for voxelized outputs, combining semantic segmentation accuracy with instance detection recall into a unified score.	[246]
⚖ **Downstream Evaluation - VQA**				
Top-1 Acc	↑	Visual Question Answering Top-1 Accuracy	Exact-match accuracy of predicted answers across diverse question categories in autonomous driving VQA tasks, measuring reasoning reliability.	[373]

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